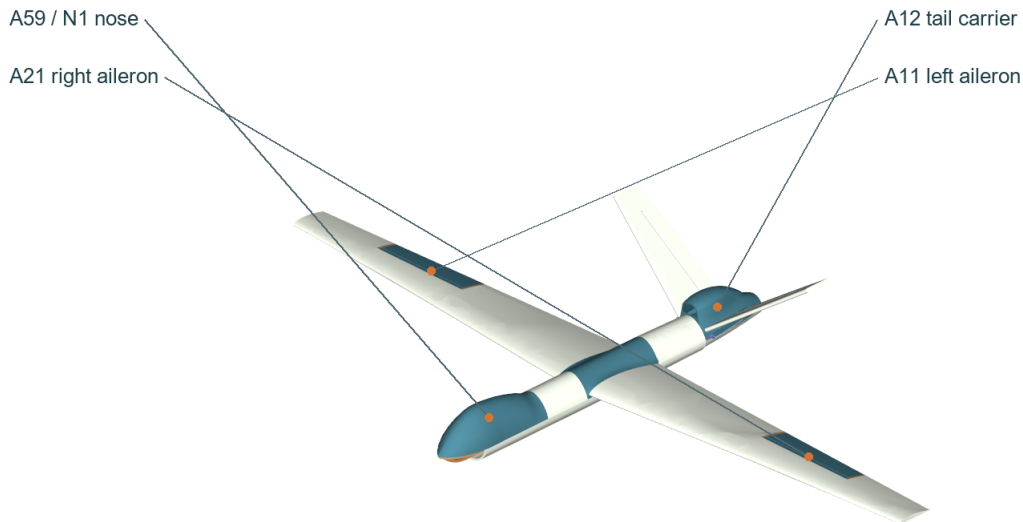


Reaper: build one stage at a time



Intended upright V-tail aircraft from actual assembly CAD. Landing gear and propeller are absent at this build stage.

DARTS founding project - registration pending. This complete follow-along guide takes the current cut parts and prints through assembly, wiring and bench checks. Landing gear is a later stage after midsems.

Use the complete current plywood cutting set, reinforced A44_L/R outer ribs and no-motor-hole A26 firewall blank. Reuse sound foamboard and prints. A11/A21 ailerons, A23/A36 fixed rails and HTRB right tail-horn backer are supplied unchanged for the missed prints.

Read the next step before bonding; for an already-built stage perform its check. Keep the propeller off throughout assembly, wiring and configuration. Physical fit, cured joints, control operation and propulsion must be proved on the aircraft before pilot acceptance.

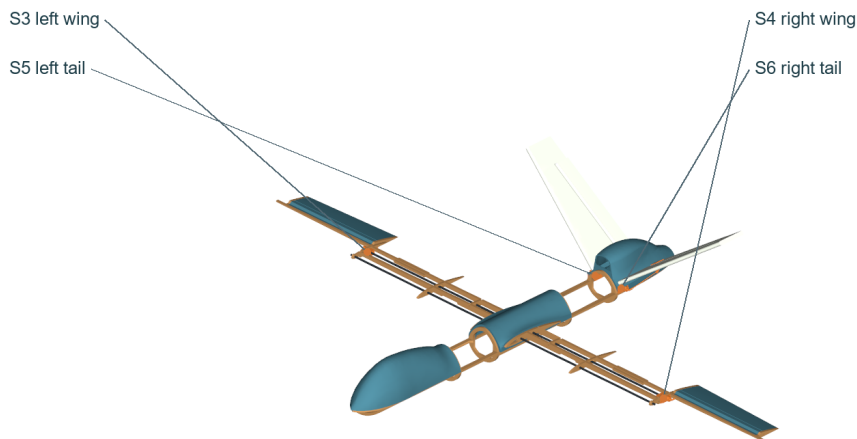
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Start with the parts you have



Right and left mean the pilot looking toward the nose. The nose is the narrow front end; the motor is at the rear. In drawings, X runs across the wings, Y increases aft and Z is upward. All quoted dimensions are millimetres unless stated otherwise.

The source wing span is 2.00 m. Two whole 5 x 1000 mm carbon rods support the wings across the centre. Plywood is nominal 6 mm and foamboard nominal 5 mm; measure your stock. Never scale parts to solve a fit.

Sort pieces into fuselage, right wing, left wing, tail and electronics bags. Match IDs to the part pictures in this guide. A40 is already incorporated into the unified A62 keel; do not stack another A40. Match A44_L/R to the reinforced plywood profiles shown here; their closed front-rod seats change insertion order and require local skin relief.

FW01/FW02 inner foam skins may be whole R5C cuts or R5E ROOT/TIP pairs. Both use the same final outline. Keep whole skins intact; join only skins already supplied as two pieces. The complete plywood recut contains all current quantities. Laser cutting makes through-profiles; the local shallow seating relief in the next pages remains a hand-finishing operation. Measure stock and a kerf/fit coupon first.

Tools, glue and curing

Use a flat board, straightedge, square, calipers, sharp knife, sanding block, clamps, hand drill, smooth trial pin, reinforced tape, zip ties, Velcro straps, soldering iron, heat-shrink and multimeter. For early servo centring use a regulated 5 V supply and standalone servo tester with neutral mode; availability is unconfirmed. A current-limited source helps the first electrical check.

For close wood-to-wood joints, use a wood adhesive approved for the plywood and clamp full contact. For print-to-wood joints, lightly abrade only the bond patch, remove dust, and qualify an adhesive explicitly compatible with both the chosen filament and wood. A gap-filling two-part epoxy is a candidate only after its offcut test passes.

For foam use an adhesive explicitly approved for its foam core and paper. Ordinary CA and solvent glue can damage exposed foam; test the exact product first. Hot glue can locate light equipment but must not be the sole battery restraint or proof of a loaded joint.

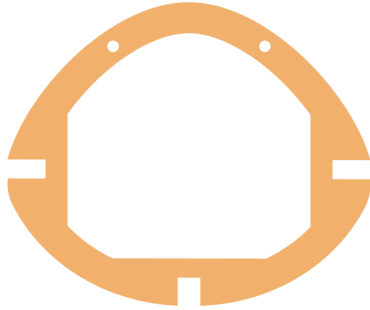
Write the exact adhesive name, mix ratio if required, working time, clamp time and full-cure time from its label on a bench card. Apply at the stated temperature. Keep each joint supported for the label clamp time and unloaded until full cure. No exact adhesive is identified in the local BOM, so there is no universal hour value to copy.

Bond one wood lap, one print-to-wood offcut and one foam fold first. After full cure pull and bend the samples. Stop if paper peels, plastic separates, the core melts or the bond remains soft; choose and test a compatible adhesive before using it on the aircraft.

Check before continuing: Continue only when the exact adhesives, clamp times and full-cure times are written down and the sample joints pass.

Fuselage wood

Match ID, shape and side. Quantities refer to installed pieces; foam cutting patterns can combine several shapes. See the foam ID table for blank counts.



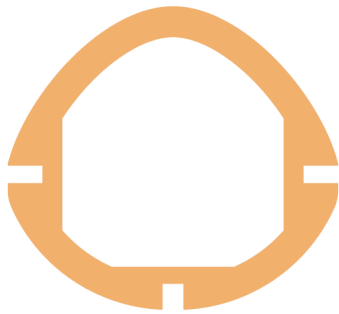
A04 - Rear body former

Plywood, nominal 6 mm; quantity 1



A13 - Front horizontal fuselage frame

Plywood, nominal 6 mm; quantity 1



A16 - Body former at Y233.65mm

Plywood, nominal 6 mm; quantity 1



A18 - Tail-carrier bulkhead - upper perimeter clearance

Plywood, nominal 6 mm; quantity 1



A25 - Forward main-gear crossmember

Plywood, nominal 6 mm; quantity 1

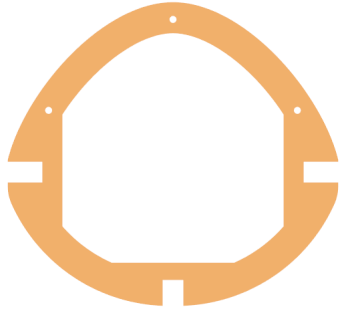


A26 - Motor firewall blank - hand drill motor pattern

Plywood, nominal 6 mm; quantity 1

Fuselage wood continued

Match ID, shape and side. Quantities refer to installed pieces; foam cutting patterns can combine several shapes. See the foam ID table for blank counts.



A34 - Front body former at Y-130mm

Plywood, nominal 6 mm; quantity 1



A43 - Aft horizontal fuselage frame

Plywood, nominal 6 mm; quantity 1



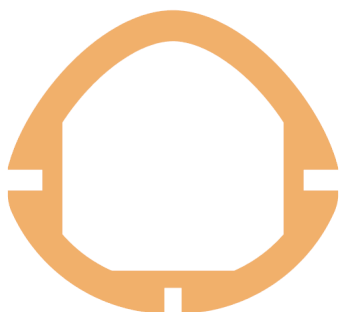
A47 - Main-gear crossplate

Plywood, nominal 6 mm; quantity 1



A48 - Aft main-gear crossmember

Plywood, nominal 6 mm; quantity 1



A54 - Body former at Y-5mm

Plywood, nominal 6 mm; quantity 1



A62_F - Lower fuselage keel - front half

Plywood, nominal 6 mm; quantity 1; length 529.07 mm

Fuselage wood continued

Match ID, shape and side. Quantities refer to installed pieces; foam cutting patterns can combine several shapes. See the foam ID table for blank counts.



A62_A - Lower fuselage keel - aft half

A62_DL - Keel joint - left side lap

Plywood, nominal 6 mm; quantity 1; length 482.65 mm

Plywood, nominal 6 mm; quantity 1; length 120.0 mm

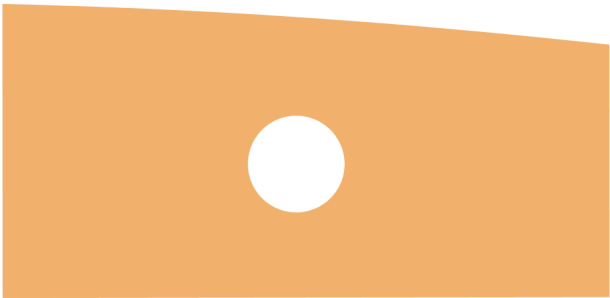


A62_DR - Keel joint - right side lap

Plywood, nominal 6 mm; quantity 1; length 120.0 mm

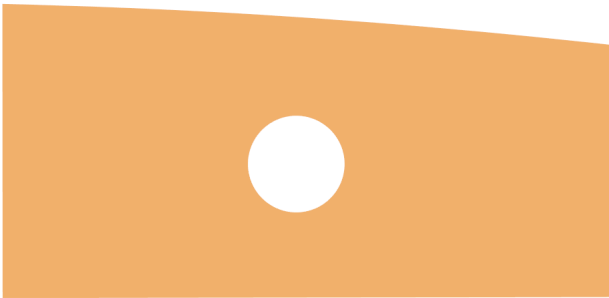
Wing wood

Match ID, shape and side. Quantities refer to installed pieces; foam cutting patterns can combine several shapes. See the foam ID table for blank counts.



A01_R - Rear-rod support rib - right

Plywood, nominal 6 mm; quantity 1



A01_L - Rear-rod support rib - left

Plywood, nominal 6 mm; quantity 1



A05_R - Front wing spar - right

Plywood, nominal 6 mm; quantity 1



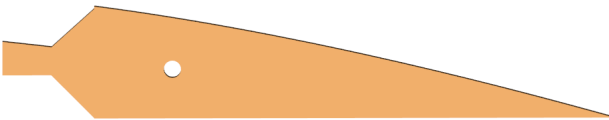
A05_L - Front wing spar - left

Plywood, nominal 6 mm; quantity 1



A10_R - Outer aileron endplate - right

Plywood, nominal 6 mm; quantity 1



A10_L - Outer aileron endplate - left

Plywood, nominal 6 mm; quantity 1

Wing wood continued

Match ID, shape and side. Quantities refer to installed pieces; foam cutting patterns can combine several shapes. See the foam ID table for blank counts.



A24_R - Middle wing rib at span250mm - right

Plywood, nominal 6 mm; quantity 1



A24_L - Middle wing rib at span250mm - left

Plywood, nominal 6 mm; quantity 1



A35_R - Inner wing rib at span53mm - right

Plywood, nominal 6 mm; quantity 1



A35_L - Inner wing rib at span53mm - left

Plywood, nominal 6 mm; quantity 1



A44_R - Reinforced outer wing rib - right

Plywood, nominal 6 mm; quantity 1



A44_L - Reinforced outer wing rib - left

Plywood, nominal 6 mm; quantity 1

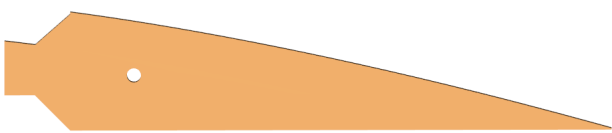
Wing wood continued

Match ID, shape and side. Quantities refer to installed pieces; foam cutting patterns can combine several shapes. See the foam ID table for blank counts.



A53_R - Inner aileron endplate - right

Plywood, nominal 6 mm; quantity 1



A53_L - Inner aileron endplate - left

Plywood, nominal 6 mm; quantity 1



A07_R_I - Rear wing spar - right inboard section

Plywood, nominal 6 mm; quantity 1



A07_R_O - Rear wing spar - right outboard section

Plywood, nominal 6 mm; quantity 1



A07_R_D1 - Rear wing spar - right first face lap

Plywood, nominal 6 mm; quantity 1



A07_R_D2 - Rear wing spar - right second face lap

Plywood, nominal 6 mm; quantity 1

Wing wood continued

Match ID, shape and side. Quantities refer to installed pieces; foam cutting patterns can combine several shapes. See the foam ID table for blank counts.



A07_L_I - Rear wing spar - left inboard section

Plywood, nominal 6 mm; quantity 1



A07_L_O - Rear wing spar - left outboard section

Plywood, nominal 6 mm; quantity 1



A07_L_D1 - Rear wing spar - left first face lap

Plywood, nominal 6 mm; quantity 1

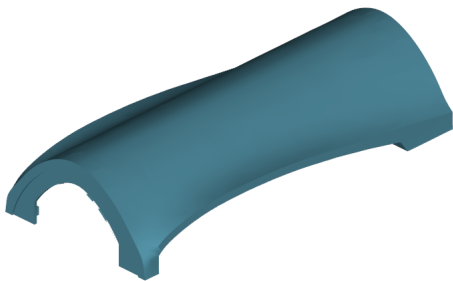


A07_L_D2 - Rear wing spar - left second face lap

Plywood, nominal 6 mm; quantity 1

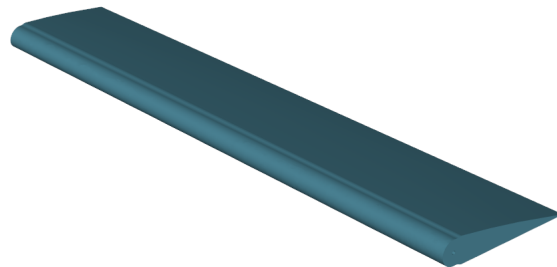
Printed aircraft parts

Match ID, shape and side. Quantities refer to installed pieces; foam cutting patterns can combine several shapes. See the foam ID table for blank counts.



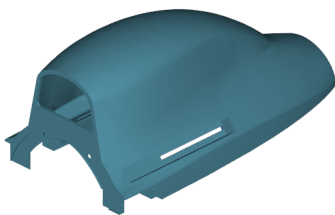
A06 - Centre fairing

PLA+ recommended / PLA owned; quantity 1



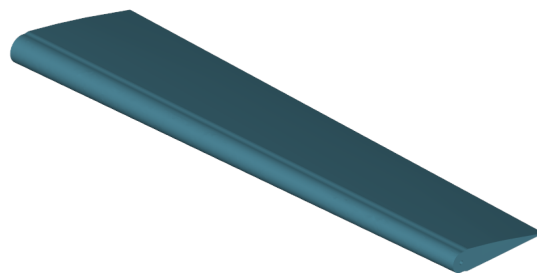
A11 - Left printed aileron

PLA+ recommended / PLA owned; quantity 1



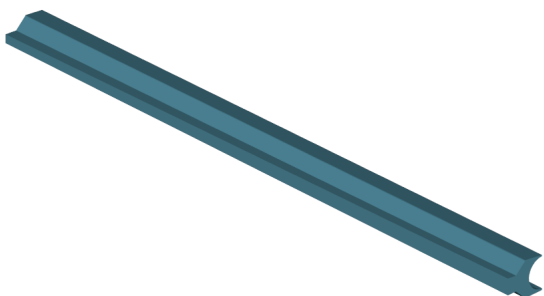
A12 - Tail carrier

PLA+ recommended / PLA owned; quantity 1



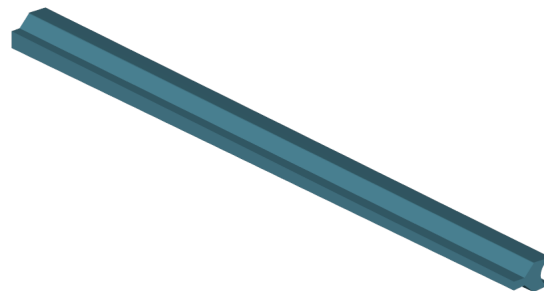
A21 - Right printed aileron

PLA+ recommended / PLA owned; quantity 1



A23 - Left fixed aileron hinge rail

PLA+ recommended / PLA owned; quantity 1

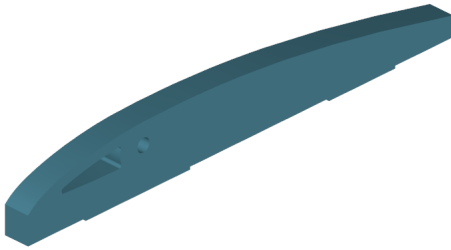


A36 - Right fixed aileron hinge rail

PLA+ recommended / PLA owned; quantity 1

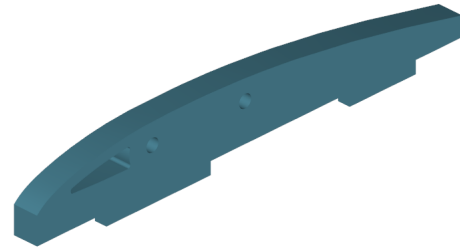
Printed aircraft parts continued

Match ID, shape and side. Quantities refer to installed pieces; foam cutting patterns can combine several shapes. See the foam ID table for blank counts.



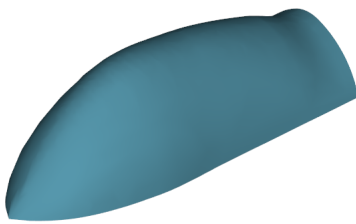
A45 - Right wing root

PLA+ recommended / PLA owned; quantity 1



A55 - Left wing root

PLA+ recommended / PLA owned; quantity 1



A59 - Owned N1 nose, reused

PLA+ recommended / PLA owned; quantity 1



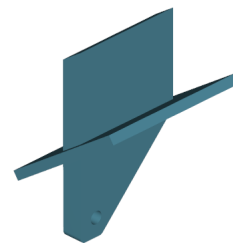
HWR - Right aileron horn

PLA+ recommended / PLA owned; quantity 1



HWL - Left aileron horn

PLA+ recommended / PLA owned; quantity 1

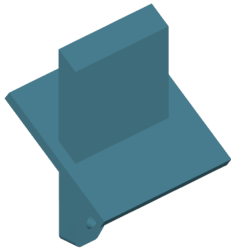


HTR - Right tail horn with base

PLA+ recommended / PLA owned; quantity 1

Printed aircraft parts continued

Match ID, shape and side. Quantities refer to installed pieces; foam cutting patterns can combine several shapes. See the foam ID table for blank counts.



HTL - Left tail horn with base

PLA+ recommended / PLA owned; quantity 1



HTRB - Right tail horn backing plate

PLA+ recommended / PLA owned; quantity 1



HTLB - Left tail horn backing plate

PLA+ recommended / PLA owned; quantity 1

Foam installed shapes

Match ID, shape and side. Quantities refer to installed pieces; foam cutting patterns can combine several shapes. See the foam ID table for blank counts.



A08 - Forward upper body foam

Foam; quantity 1



A15 - Aft upper body foam

Foam; quantity 1



A17 - Middle lower body foam

Foam; quantity 1



A20 - Tail lower body foam

Foam; quantity 1



A22 - Left fixed V-tail foam

Foam; quantity 1



A27 - Left outer wing foam

Foam; quantity 1

Foam installed shapes continued

Match ID, shape and side. Quantities refer to installed pieces; foam cutting patterns can combine several shapes. See the foam ID table for blank counts.



A38 - Right outer wing foam

Foam; quantity 1



A41 - Right moving V-tail foam

Foam; quantity 1



A42 - Aft lower body foam

Foam; quantity 1



A49 - Right inner wing foam

Foam; quantity 1



A52 - Left inner wing foam

Foam; quantity 1



A56 - Forward lower body foam

Foam; quantity 1

Foam installed shapes continued

Match ID, shape and side. Quantities refer to installed pieces; foam cutting patterns can combine several shapes. See the foam ID table for blank counts.



A57 - Right fixed V-tail foam

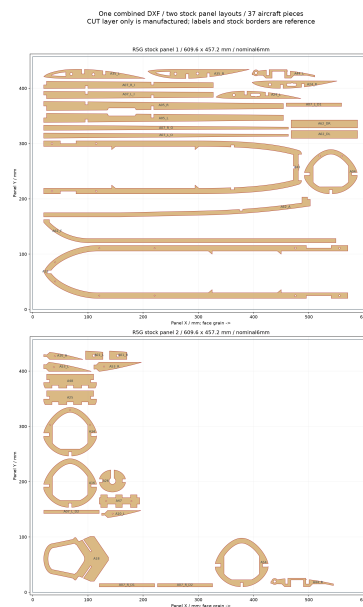
Foam; quantity 1



A60 - Left moving V-tail foam

Foam; quantity 1

Cut the complete plywood set on two stock panels



Both nominal stock layouts in the same file. Reference sheet borders and labels are not laser cuts.

Reaper_ALL_PLYWOOD_R5G_6mm_PROVISIONAL.dxf contains all 37 aircraft wood pieces at quantity 1 each, including mirrored pairs, four rear-spar sections and four face laps, both keel halves/laps, the reinforced A44 pair and no-motor-hole A26. The identification cards show every current part.

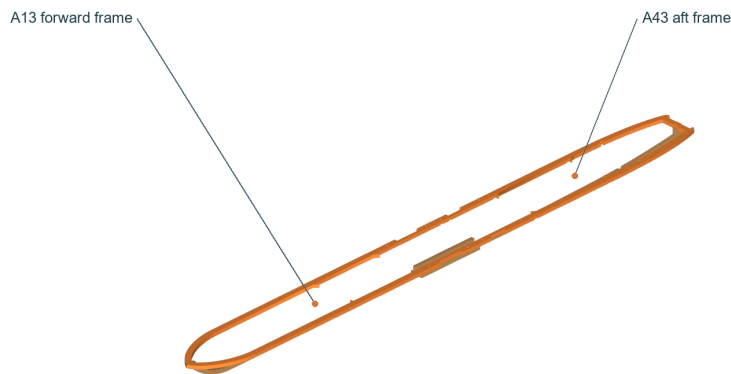
One combined DXF contains two separate 609.6 x 457.2 mm stock layouts, nominal 6 mm thick. Stock 1 starts at global X0; stock 2 starts at X659.6. The full drawing width is 1269.2 mm. If the laser bed does not span this width, process each stock region as a separate job on its own sheet without rescaling.

Cut only the CUT layer. Disable REFERENCE_NOT_CUT stock borders and LABELS_NOT_CUT text in the machine job. Preserve the nest rotations and grain direction: the strongest face grain follows each stock panel X axis. Labels and borders are placement references.

Actual stock thickness, grain and kerf are unmeasured. Qualify the critical rod-hole and rib/spar/keel fit coupons before cutting the full set. Kerf compensation is separate from modeled A18 perimeter clearance. After cutting, hand-finish local shallow seats and the sacrificial A44 tail underside to the shown finished taper. Drill hardware holes only against actual parts.

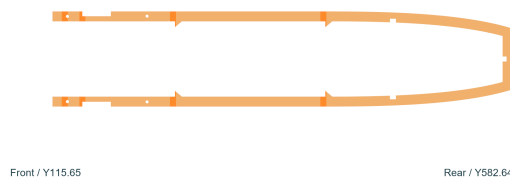
Check before continuing: Two real stock sheets and machine job regions are confirmed; CUT-only selection, scale, grain and fit coupons pass.

Finish only the marked wood seating patches



Lower: actual removal shapes identify four local A43 seating regions. Orange is material removed, never an extra part or a whole-frame thinning instruction.

A43 underside - orange is removed only at the four local seats



Front / Y115.65

Rear / Y582.64

Dry-fit complete pieces before glue. Nominal stock is 6 mm; actual thickness and laser kerf remain unmeasured. Through-cut outlines and shallow hand-finished seats are separate operations. Never thin a complete frame to fix a local seat.

On A43 remove only the orange source removal volumes at Y125-131, Y142.7-148.7, Y233.65-239.65 and Y384.65-390.65. The first two flat underside seats lose 1 mm, leaving 5 mm. Rear seats follow their curved removal geometry: a representative residual is about 4.83 mm; the complete bounding box is not a uniform 5 mm rebate.

A05_L/R and A07_L_I/R_I need local rib-tab relief near X53-59 and X247.35-253.35 and their mirrors. A62_F has shallow A54-seat relief at Y-5 to 1, leaving a 5 mm keel width locally. Use the actual mate and marked removal shape as the stop.

Dress tiny A53/rear-spar wedges only where they prevent seating. Make small cuts, refit and stop when each joint seats without force. Preserve the surrounding 6 mm structure; measure any defect outside the shown patches before removing wood.

Check before continuing: All mates seat without force; each relief remains within its marked patch.

A43 underside seats and curved rear relief

A43 underside - orange is removed only at the four local seats

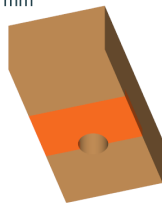


Front / Y115.65

Rear / Y582.64

Top: actual A43 underside shows the marked bands on both rails. Lower: representative right-rail close-ups of the four source removal regions.

Y 125.00 to 131.00 mm



1 mm underside removal; 5 mm remains

Y 142.70 to 148.70 mm



1 mm underside removal; 5 mm remains

Y 233.65 to 239.65 mm



Curved source removal; refit actual former

Y 384.65 to 390.65 mm



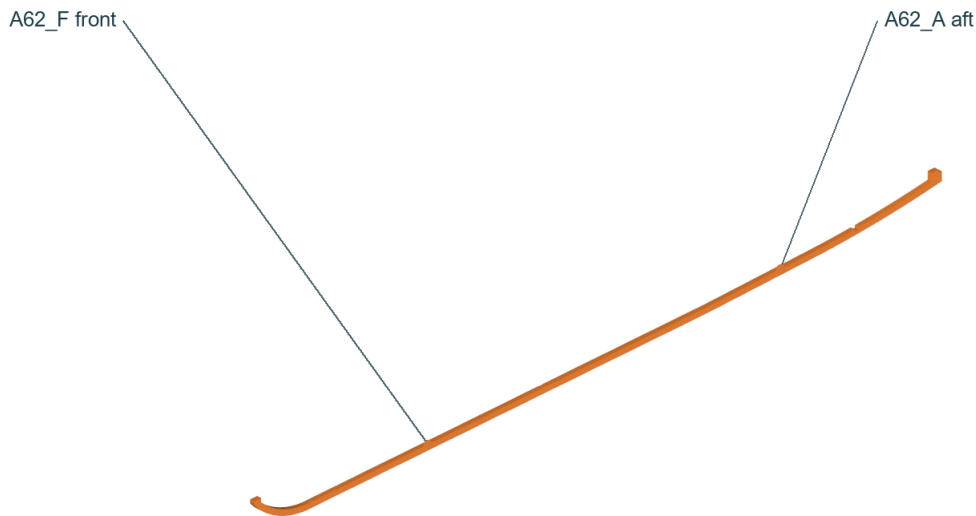
Curved source removal; refit actual former

View the underside. Orange outlines are the exact source removal solids, confined to the four marked fore-aft bands. At the flat A25/A48 seats pare 1 mm from the underside, leaving 5 mm from the Z0 top face.

The A16/A04 reliefs follow curved contacts. Their removal solids span different depths; about 4.83 mm is a representative remaining section, not a whole-region thickness. Refit the actual former progressively. Keep all wood outside the highlighted removal shapes at its nominal 6 mm.

Check before continuing: Mark only highlighted contact patches; calipers and real mates confirm seating without full-frame thinning.

Align the two keel halves



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

Take A62_F (front keel) and A62_A (aft keel). Front length is 529.070 mm; aft finished length is 482.650 mm. Stand their broad side faces in one plane on a straight jig, matching the top and bottom edge shapes. The butt seam is at Y100.

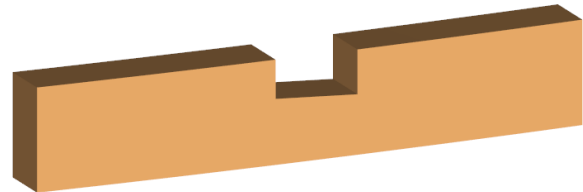
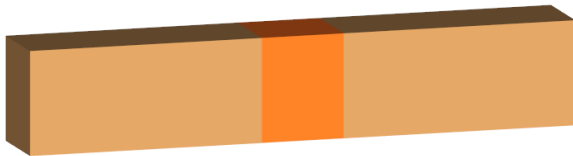
Use the A13/A43 frames loosely as an alignment check before bonding. Do not use glue to fill a misaligned step. The seam locates the halves; the two side laps in the next step carry the joint.

Check before continuing: A straightedge crosses the seam without a step or kink, and the dry frames seat.

Keel lengths and local notch finishing

A54 seat: Y-5 to 1; local width 5 mm

A18 notch: Y454.5 to 460.5; floor Z-31



Close-ups use actual placed keel geometry and the A54 source removal solid. The aft notch floor remains a seating surface.

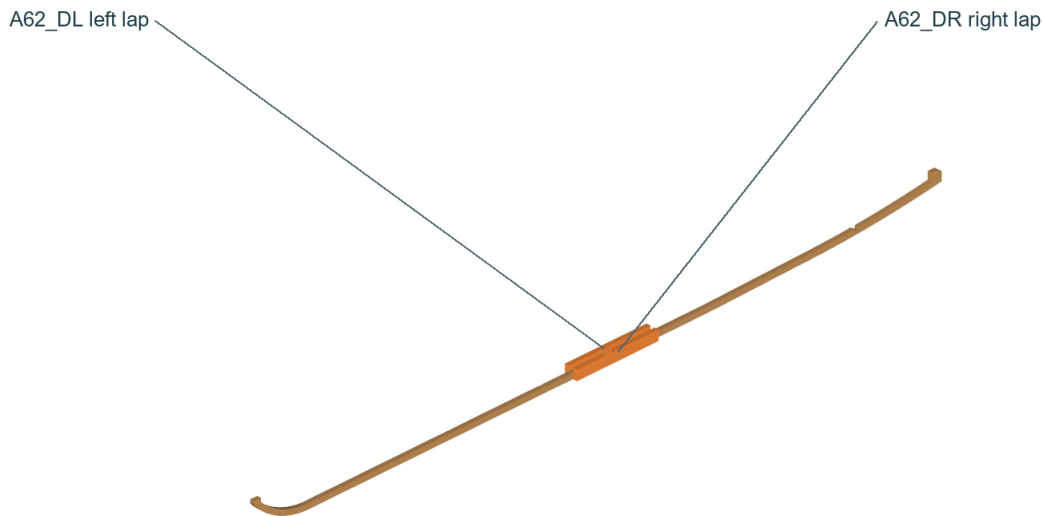
The front keel runs Y-429.070 to 100, length 529.070 mm. The aft keel runs Y100 to 582.650, length 482.650 mm. The complete current cut set already shortens the old aft projection by 13.015 mm. Keep the butt seam at Y100 and side laps at Y40-160.

At A54, Y-5 to 1, the front keel has shallow side relief to a local 5 mm width. This partial-depth operation cannot be laser through-cut. Follow the actual removal solid and mate; keep the neighbouring stock at 6 mm.

At A18, Y454.5-460.5, preserve the aft notch floor at Z-31. The 5 mm bottom tongue bears on that floor. Nominal keel notch and slot fit do not supply measured stock clearance; trial a representative fit coupon before bonding.

Check before continuing: Both halves and laps align; local front relief and aft notch seating pass on real wood.

Bond both keel side laps



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

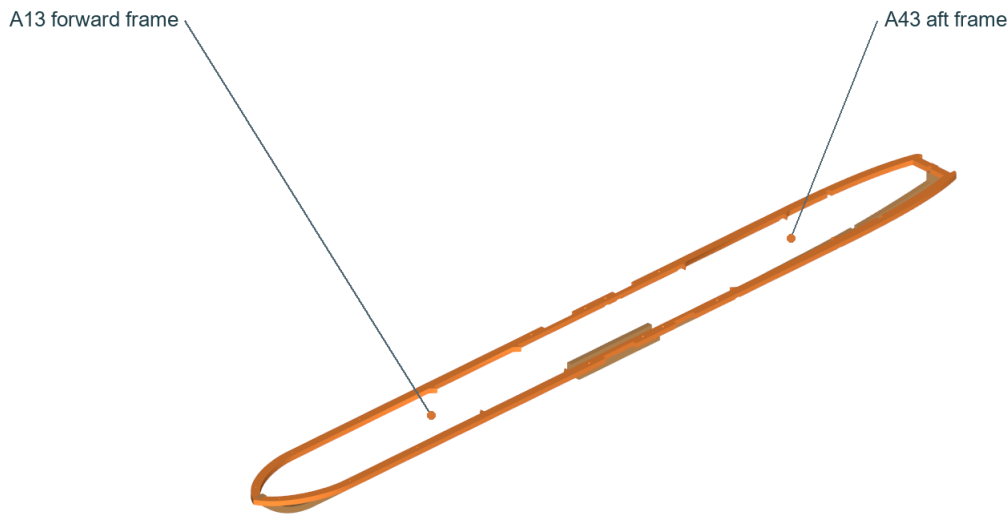
Fit A62_DL to the left broad side and A62_DR to the right side of the aligned keel. Each 120 mm lap spans Y40-160, giving 60 mm on each half. Align their lower edges with the keel at Z-41.714.

Apply the qualified wood adhesive to the complete mating faces. Clamp the halves and both laps flat in the jig. Wipe squeeze-out away from nearby slots. Keep the jig in place through clamp time and leave the joint unloaded through full cure.

If the keel is already joined, inspect both laps, the straight seam and the cured contact. Do not dismantle a sound joint to repeat this step.

Check before continuing: Both opposite laps are installed, fully bonded and straight; no loose edge, soft glue or split wood.

Dry-fit the horizontal body frames



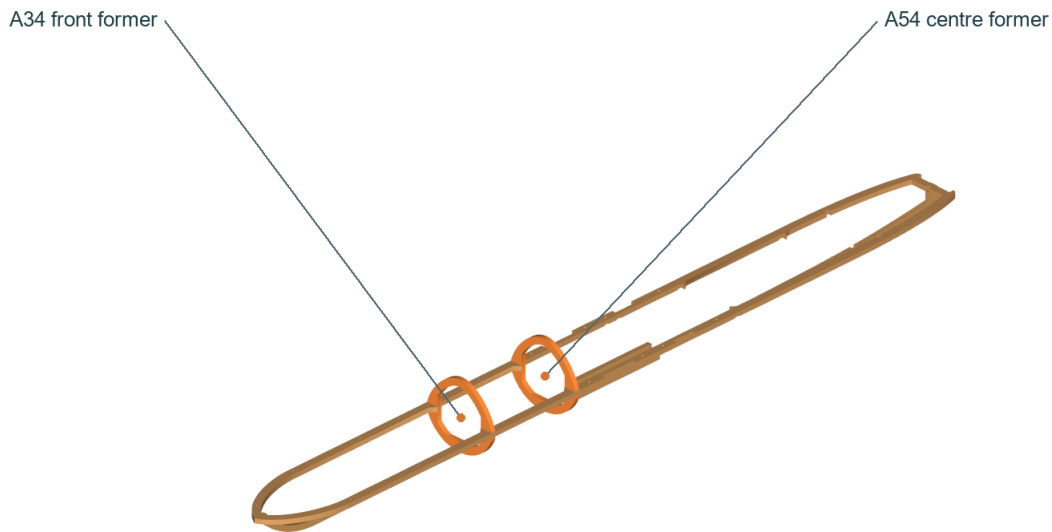
Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

Fit A13 at the front and A43 at the rear onto the keel tabs. Their upper faces are the common Z0 plane. The rounded front of A13 points toward the nose; A43 runs toward the motor.

Keep the assembly supported so the thin frame members do not bow. Fit without glue now; the formers, gear-bay pieces, nose and tail carrier must all agree before you bond the body.

Check before continuing: The frames sit level on a straight keel, all tabs seat and the structure stays square without force.

Add the two forward body formers



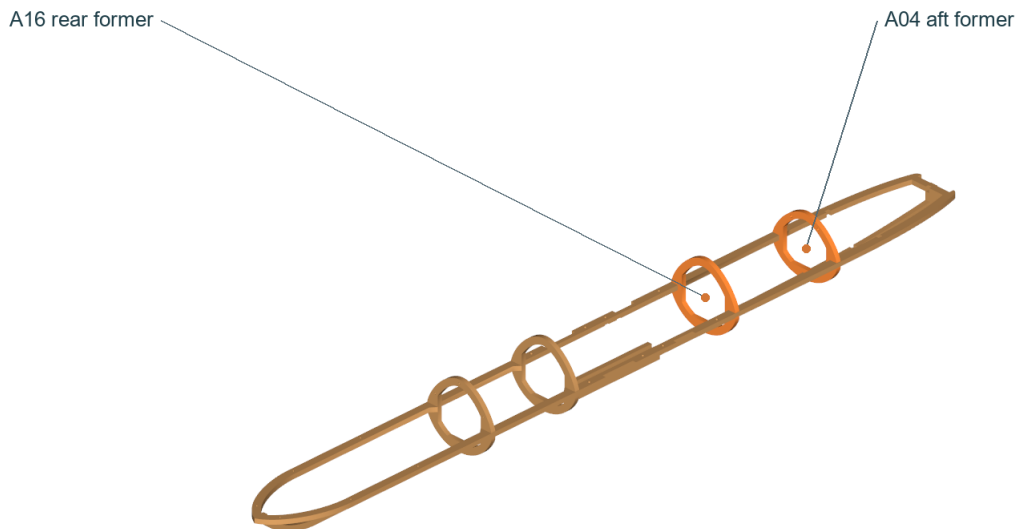
Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

A34 is the front ring former, with its forward face at Y-130. Fit its lower slot and frame contacts first. A54 is the centre ring former, with its forward face at Y-5: 125 mm aft of A34.

The former rings stand upright across the body. Keep their top arcs above the frame and their lower notches toward the keel. Check a square against each ring. Use the local A62_F relief from Step 1 if A54 will not seat.

Check before continuing: The two rings are upright, square to the centreline and 125 mm apart between forward faces.

Add the two aft body formers



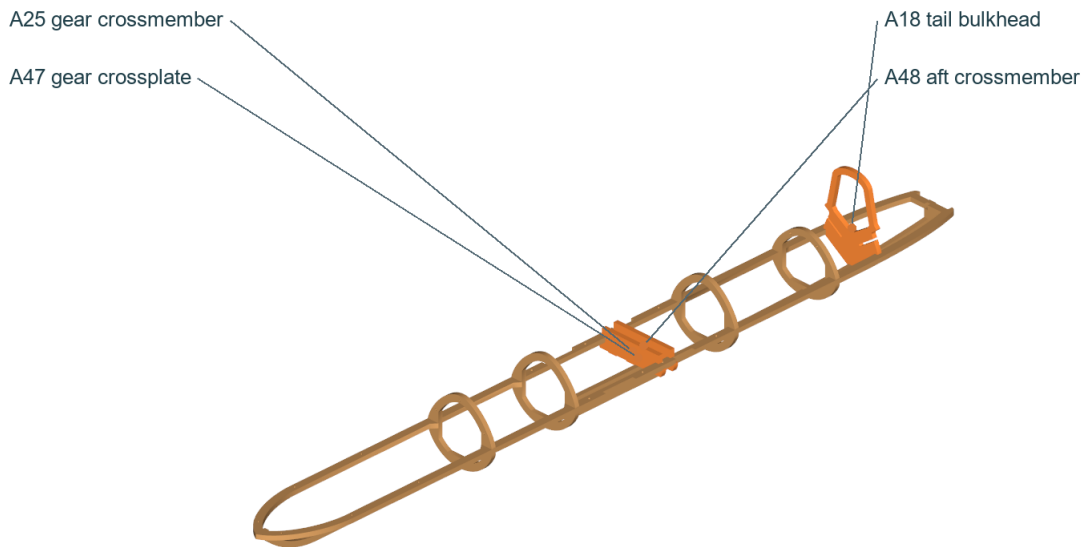
Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

Fit A16 with its forward face at Y233.65, then A04 with its forward face at Y384.65. From the A34 forward face these distances are 363.65 and 514.65.

Keep each ring square and the top of A43 at Z0. If a former lifts A43, use only the documented local underside seat relief. Trial a lower foam wrap over the rings to see whether the body outline remains smooth.

Check before continuing: All four ring formers seat without twisting the body; left and right frame rails remain level.

Fit the gear-bay pieces and tail bulkhead



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

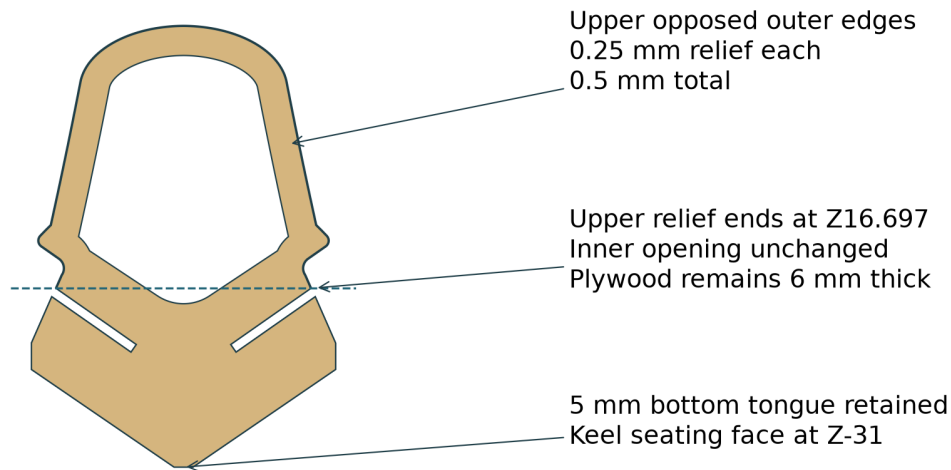
Dry-fit A25 at Y125 and A48 at Y142.7, then A47 across their matching tabs. These three pieces stay in the wood skeleton even though purchased landing gear is deferred until after midsems.

Fit A18 tail-carrier bulkhead with its aft face at Y460.5. Its central opening clears the keel while its 5 mm bottom tongue seats on the keel notch floor at Z-31. Above Z16.697, the opposed outer receiver edges have 0.25 mm relief each, 0.5 mm total. The plate remains 6 mm thick; preserve the lower tongue and notch floor.

Leave the motor mount out until its separate fit step. Existing gear-bay structure is not proof of a bolt pattern or load capacity for the later gear.

Check before continuing: A25/A47/A48 sit flat and A18 is square; later gear and motor access remain open.

Identify the tail-carrier receiver before glue



Actual A18 front view: the upper opposed external edges receive 0.25 mm relief per side. The lower tongue, inner opening and keel seat remain.

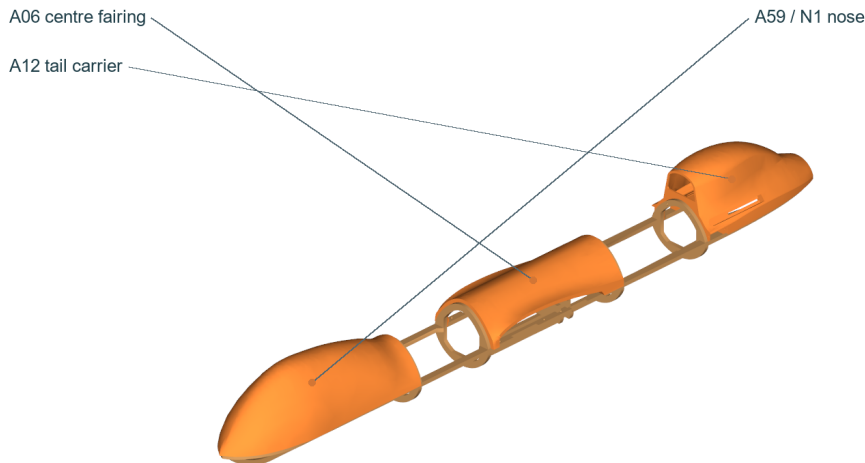
A18 is nominal 6 mm plywood. Its 0.5 mm total change is a perimeter allowance: 0.25 mm inward on each opposed upper external edge above Z16.697. The inner opening, lower arms, 5 mm bottom tongue and Z-31 seating face are unchanged.

The current adapted A12 file has a 6.3 mm axial receiver slot, nominally 0.15 mm per side of a 6 mm plate. This is separate from the upper perimeter relief. An older unadapted print has a local 2.5 mm slot and cannot take a 6 mm plate by contour relief alone. Measure the actual printed receiver and plywood.

The source matches when installed, but simple forward, aft, up and down insertion/removal movements are obstructed. Dry-fit the actual tilt/flex/order with the body open, without forced bending, before bonding A18 or A12. A successful installed CAD fit does not establish a removable shell.

Check before continuing: Actual print version, measured slot width and safe physical assembly order are established before glue.

Trial the printed body parts, then bond the wood



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

Trial the owned A59/N1 nose, A06 centre fairing and A12 tail carrier against their frames before glue. A18/A12 have a clear installed fit in CAD, but straight insertion/removal is obstructed. Establish the real tilt/flex/assembly order with the body open before bonding either part. Stop if the receiver cannot seat without damage; do not force a shell over proud wood tabs.

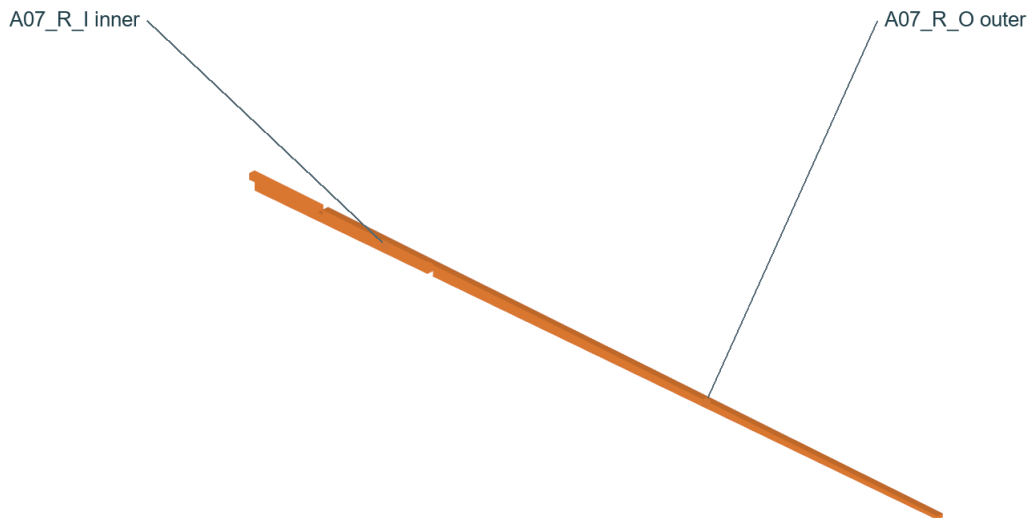
The prints attach at their actual wood seating faces and sockets. A12 surrounds the tail bulkhead; A06 follows the centre formers and frame; A45/A55 roots seat against the wing structure and locate the two carbon rods. These contact patches carry load into the skeleton. Foam skin and tape can close seams but cannot replace the sound wood-to-print joints.

Mark which print contacts need the qualified print-to-wood adhesive. For the nose and access covers retain a removable path using tape/ties at sound frame contact, rather than sealing electronics inside. There is no dimensioned universal screw pattern for all shells.

When the complete dry body aligns, remove shells, bond wood contacts and clamp square through label clamp time and full cure. Then bond A12 to its A18 seating faces with the qualified print-to-wood adhesive and hold square until fully cured. Retain A06 serviceably with tape/ties at sound frame contact. Keep the nose removable, the motor mount out and the body open.

Check before continuing: The cured wood is straight; shells seat by hand and motor, battery and USB/SD access can be reached.

Align each rear wing spar



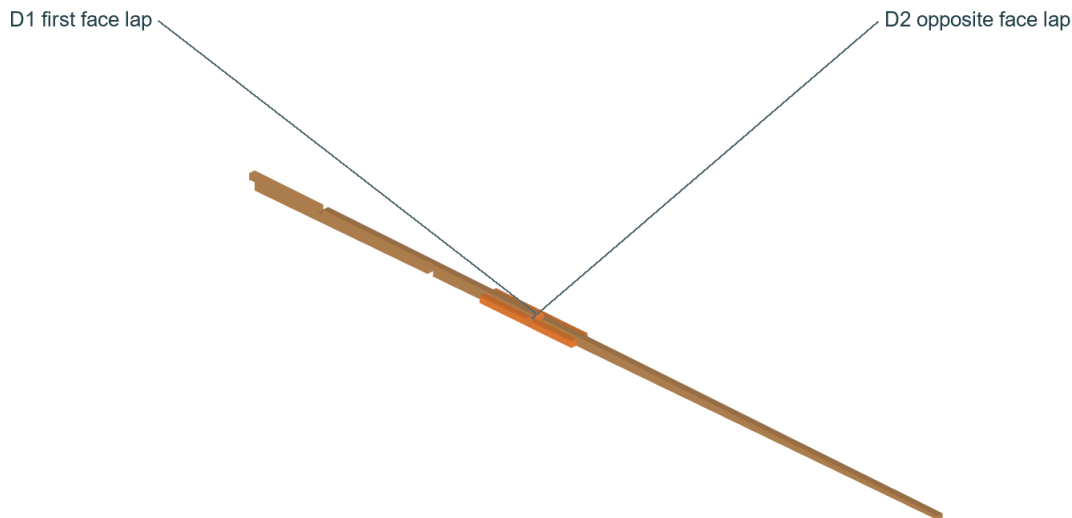
Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

On the right, align A07_R_I (inner section) with A07_R_O (outer section). Their seam is at X360. On the left use A07_L_I and A07_L_O, with the mirrored seam at X-360.

Lay each spar broad face on a flat jig. Keep its notches facing the matching ribs. Trial-fit the ribs before glue so a reversed half is caught now.

Check before continuing: Each spar is straight across its seam and its notches match the correct-side ribs.

Bond both face laps on each spar



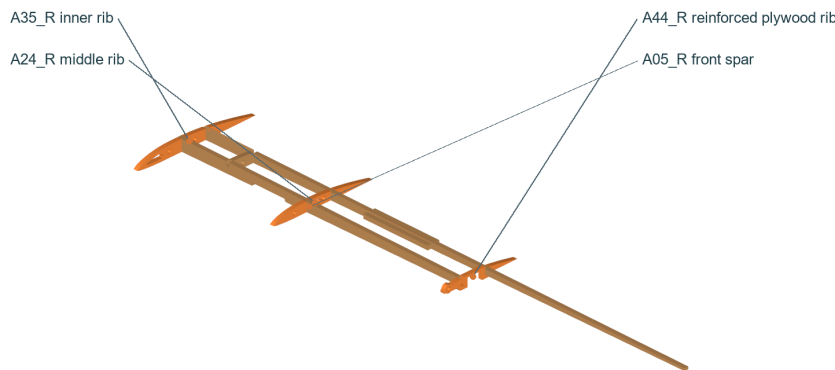
Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

Fit A07_R_D1 and A07_R_D2 to opposite broad faces of the right spar. Each 100 mm lap covers X310-410, giving 50 mm per half. Repeat with A07_L_D1/D2 on the left.

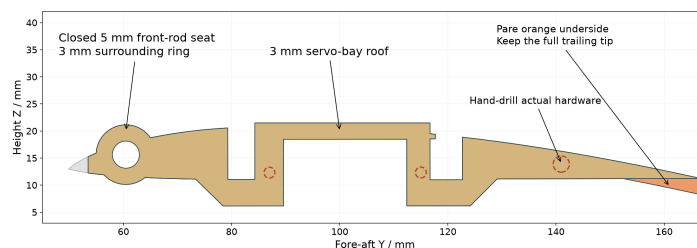
Bond full contact with the qualified wood adhesive. Clamp straight and keep glue out of rib notches. Both laps are structural pieces; a rigid-looking butt joint is not a reason to omit one. Leave unloaded until full cure.

Check before continuing: Both spars have two opposite cured laps, straight edges and open rib seats.

Build the right wing with its reinforced plywood rib



Current reinforced plywood outer rib in the actual wing frame. Lower detail shows its closed front seat and 3 mm roof; rods enter axially before skin closure.



Lay out A05_R front spar, joined A07_R rear spar, A35_R inner rib, A24_R middle rib and A01_R rear-rod support rib. Use the current reinforced nominal 6 mm plywood A44_R at the outer station.

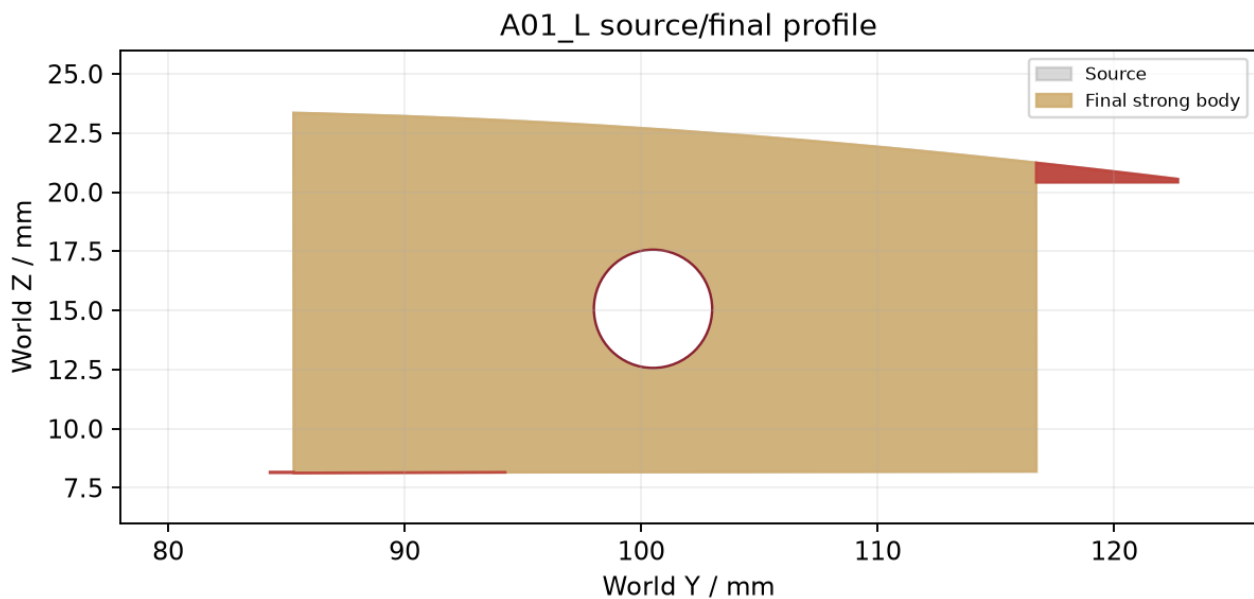
Ribs run fore-aft, rounded noses toward the leading edge: inner station about X53, middle X250 and outer X490. A01_R supports the rear rod near X130. Its tiny lower front toe and upper rear shelf are removed; fixture A07 at its original height while bonding the broad spar contact faces.

A44 has a 3 mm servo-bay roof and a closed front-rod seat with a 3 mm surrounding ring. The leading fine tip is shortened 3.632 mm; keep the complete original trailing tip. Laser stock adds a sacrificial underside tail band to maintain a 3 mm cutting web. Pare only that underside allowance to the finished source taper before fitting A53 or the skin. Trial the actual servo and skins; relieve upper and lower inner foam locally near the closed front seat on each wing. Keep outer paring intact.

Keep the full trailing tip and its A53 contact footprint; never shorten it to remove the sacrificial band. The three inherited small A44 holes are omitted from the laser CUT profile. Mark and hand-drill only after the actual servo tabs/retainer fit is established; drilling can recreate thin webs. Dry-fit the whole frame and rod path, then bond the full wood contacts with tested wood adhesive and cure without twist.

Check before continuing: Spar height and airfoil attitude match the jig; servo access, local skin relief and axial rod path are clear.

A01 rod support and spar-height jig



Actual final A01 profile and mating spar contacts. The tiny removed toe/shelf no longer provides a height stop.

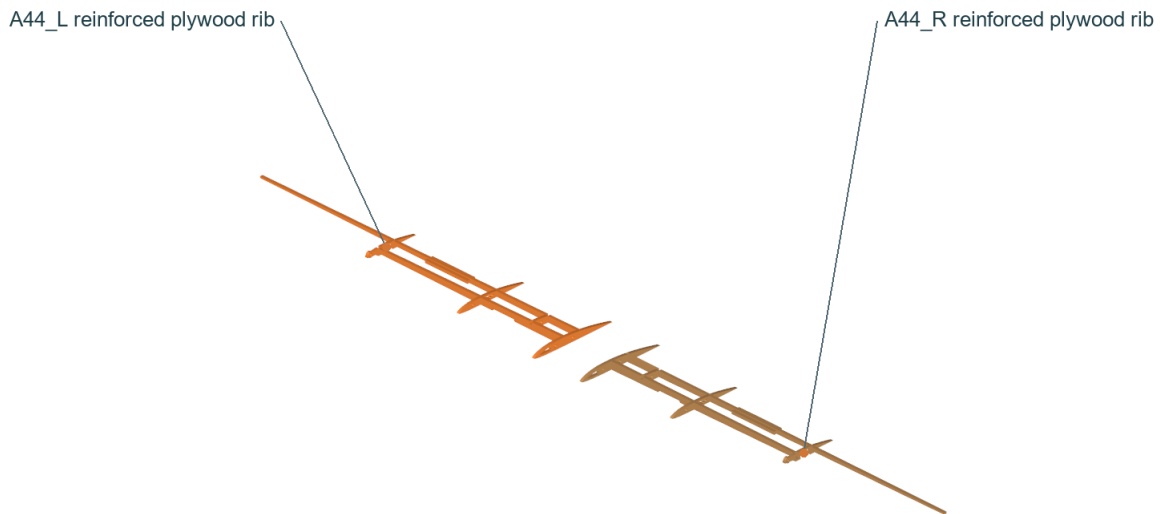
A01_L/R retain the robust body and exact 5 mm rear-rod hole at Y100.500, Z15.070. Only the very small 0.036 mm lower front toe and 0.141 mm upper rear shelf are removed along the existing spar contact planes.

The rear shelf no longer registers A07 height. Keep the rear spar in the airfoil jig and bond the broad vertical contact faces: about 64.677 mm² against A05 and 50.662 mm² against A07. The rod wall remains about 4.408 mm; this change does not add a skin bulge or require a new foam recess.

Dry-fit A01 from above into the open frame, then feed the rear rod axially before closing the skins. Preserve its source hole and spar contacts; a 5 mm nominal rod needs actual-fit confirmation.

Check before continuing: Rear spar stays at the jig height through cure, broad faces bond and the actual rod feeds axially.

Build the left wing and compare both sides



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

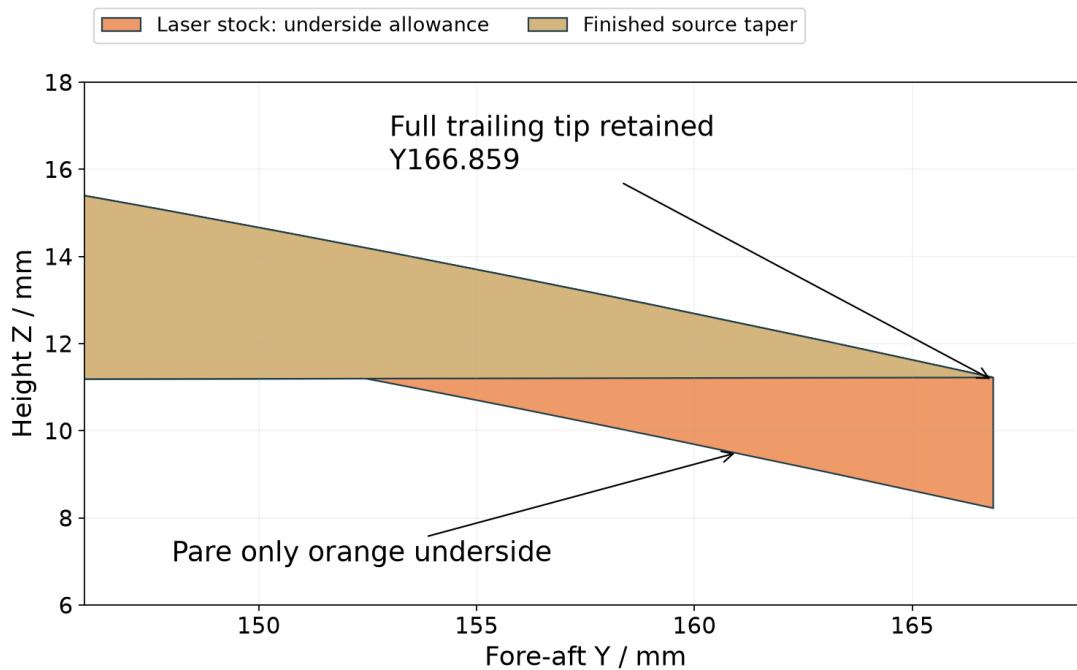
Repeat Step 11 with A05_L, A07_L, A35_L, A24_L, A01_L and reinforced plywood A44_L. This is a mirror. Match the servo pockets, closed rod seats and control attachments to their correct sides.

Compare both frames on the same flat jig before adhesive cures: rib stations, leading edges, spar joints, root faces and twist. A01 keeps its exact 5 mm rear-rod hole; its trimmed shelf no longer sets spar height, so retain the jig through cure.

Trial each outer skin and fixed hinge rail. Relieve upper and lower inner foam only where reinforced A44 material touches them near the front seat, on both wings. Preserve outer paper and the matching airfoil attitude; do not shorten one wing to conceal a mistaken fit.

Check before continuing: Both wings have matching rib spacing, spar height, airfoil attitude and clear axial rod insertion.

Pare the A44 tail underside before assembly



Actual stock and finished tail profiles. Orange is the exact secondary removal allowance; brown is retained. The full Y166.859 trailing tip remains.

The full-length trailing tip carries the source A53 contact and matching skin taper. The laser blank extends below that thin taper to keep at least a 3 mm vertical web during cutting. This extra underside wood is temporary cutting stock.

With the rib supported flat, transfer the finished side profile, then pare and sand only the orange underside allowance between approximately Y152.458 and Y166.859. Refit against the actual A53/skin progressively. Keep the complete trailing tip; do not shorten it or remove the upper foil surface.

Finish both mirrored ribs before bonding A53, fitting skins or closing the wing. The closed front-rod seat, 3 mm surrounding ring, 3 mm servo-bay roof and manual small-hole drilling remain part of the same rib.

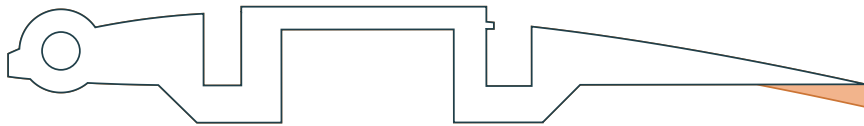
Check before continuing: Both underside tails match the finished taper; full tip, A53 contact and skin fit remain intact before bonding.

A44 paring templates at actual size

Print only this page at 100% / Actual Size. Measure the 25 mm bar before tracing; do not use Fit to Page. The two profiles are 113.394 mm long, from Y53.465 to Y166.859.

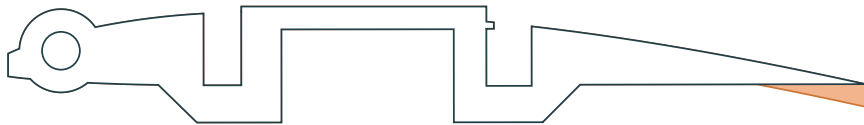
Black is the finished outline; orange underside is temporary laser stock to pare away. Preserve the full trailing tip. These thin finished profiles are hand-finishing templates, not laser cutting patterns.

A44_L - 6 mm plywood; full trailing tip retained



Finished length 113.394 mm. Trace underside taper; do not shorten the tip.

A44_R - 6 mm plywood; full trailing tip retained

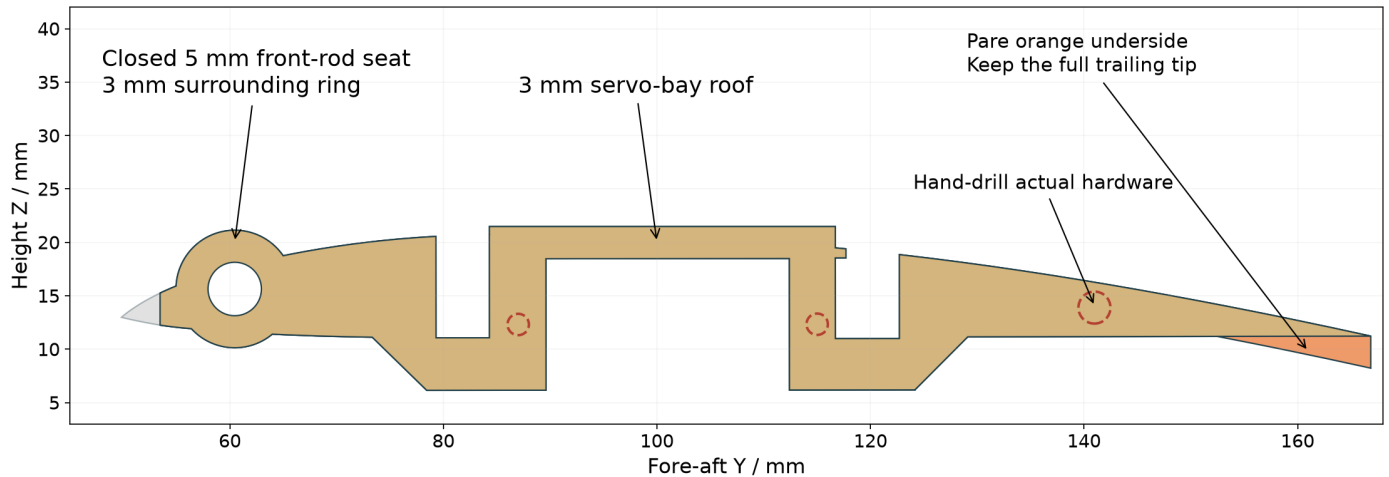


Finished length 113.394 mm. Trace underside taper; do not shorten the tip.



25 mm calibration bar - confirm with a ruler on the actual print.

Reinforced A44 seats and skin clearance



Current right-rib profile: orange tail underside is sacrificial laser stock, brown is the finished taper. Preserve the full trailing tip. Dashed small-hole datums are hand-drill references; the left rib is mirrored in installed thickness.

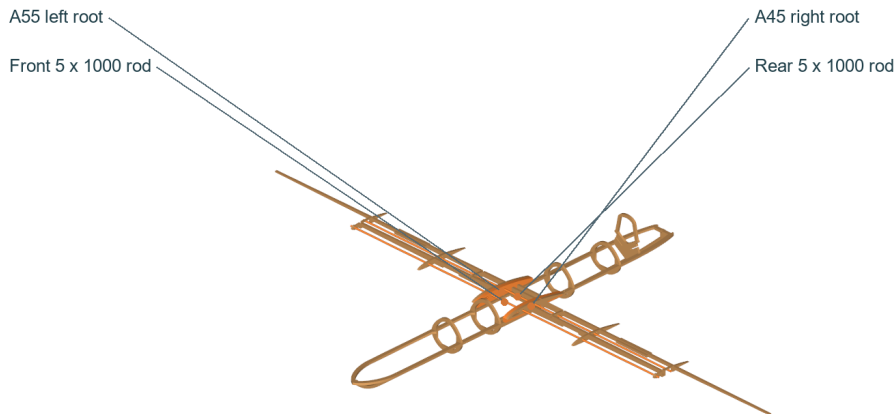
Use the reinforced nominal 6 mm plywood pair. The front seat is closed around the source 5 mm rod axis; it has a 3 mm surrounding ring. The servo-bay roof is also 3 mm. Only the leading fine tip is shortened 3.632 mm. The original trailing tip is retained to Y166.859, giving 113.394 mm fore-aft length. The laser blank has sacrificial underside tail stock; pare that allowance to the finished taper before A53 or skin fitting.

Both wings need local upper and lower inner-foam relief near the front rod seat where reinforced material reaches the skin. Preserve outer paper. Insert the front rod axially through the closed seats before closure, after the wing/root dry-fit.

Three inherited small hardware holes are omitted from laser cutting. Mark and manually drill against the actual servo tabs/retainer only after checking edge distances. An omitted hole is not proof that any chosen screw fits or that thin wood restored by drilling will carry load.

Check before continuing: Closed seat, 3 mm roof, both skin reliefs, axial rod path and actual servo retention pass before closure.

Fit both roots and the two whole carbon rods



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

Fit A55 left root and A45 right root with the complete body and both wing frames supported. Slide the front rod through the nominal 5.2 mm printed bores and mating wood seats at Y60.454/Z15.632. Slide the rear rod at Y100.500/Z15.070.

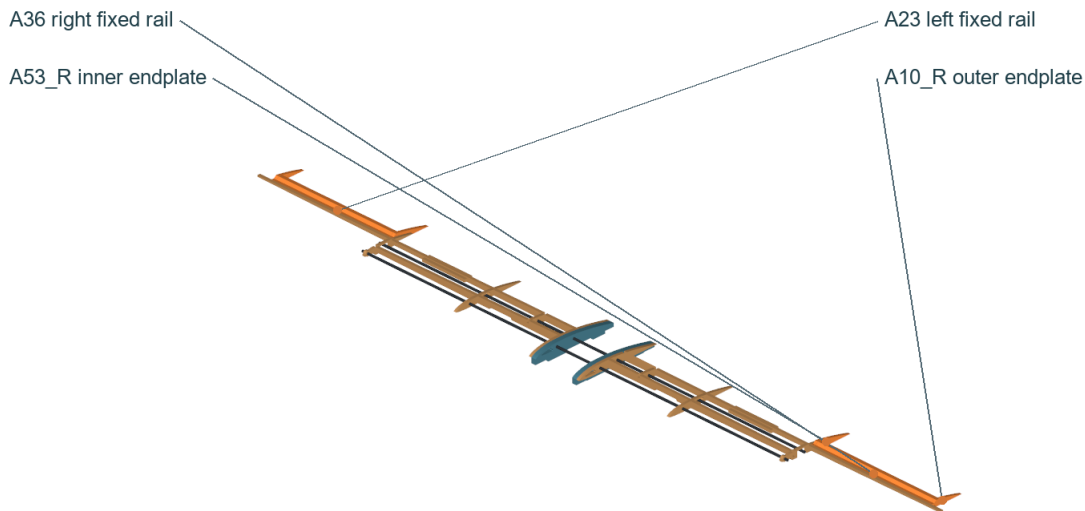
Each rod is solid carbon, 5 mm diameter and 1000 mm long, centred from X-500 to +500. Keep both whole; do not cut or butt-join them at the body. The reinforced A44 front-rod seat is closed. Insert the rod axially before outer-wing skin closure; it cannot be dropped into the seat from above. Keep both rod paths accessible during dry-fit.

Trial the actual rods, roots and wood simultaneously. Finish a bore locally only enough for a snug fit; do not scale a root. Mark the loaded root-to-wood contact and bond only after a qualified offcut joint and full dry fit.

Before covering, prove that neither a wing nor a rod can move out spanwise. The source does not define a fully qualified removable locking system. A friction fit alone is insufficient: hold this stage for a documented positive restraint and physical pull/load check if the installed bonded arrangement cannot resist escape.

Check before continuing: Rods remain whole; both roots seat without cracks; wings and rods have demonstrated restraint against escape.

Install the fixed aileron rails and endplates



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

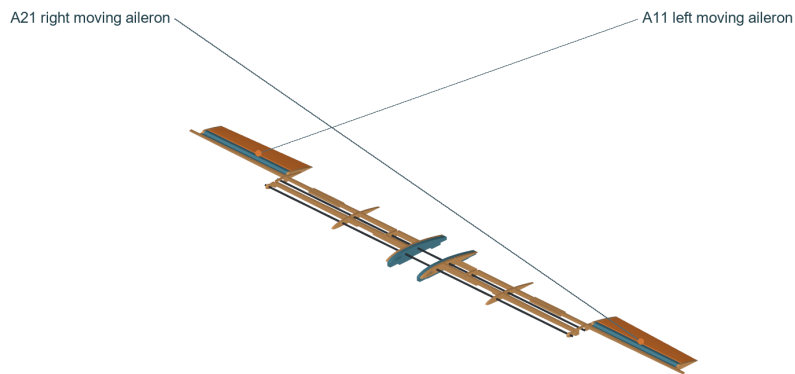
Right side: fit A36 fixed hinge rail with A53_R inner wooden endplate and A10_R outer endplate. Left side: A23 with A53_L and A10_L. Trial the corresponding moving aileron before bonding the fixed pieces.

The endplates lie beyond the moving surface ends, not inside its free span. Preserve their pivot-support material. The inner A53 datum remains about X493 on the right and its mirror on the left. Keep the printed rail aligned with the common hinge line.

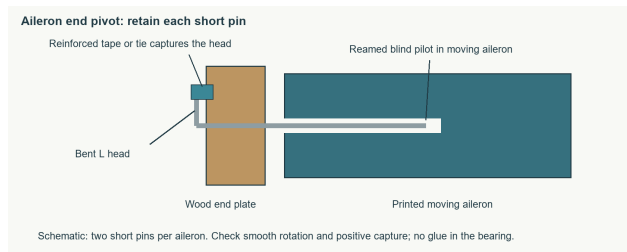
Use qualified wood adhesive at wood contacts and qualified print-to-wood adhesive at the fixed rail contact. Keep all hinge pilots free of glue and leave space for the captive pin heads.

Check before continuing: Both rails are fixed and straight; moving ailerons can fit between their endplates without being pinched.

Fit the unchanged ailerons and captive pins



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.



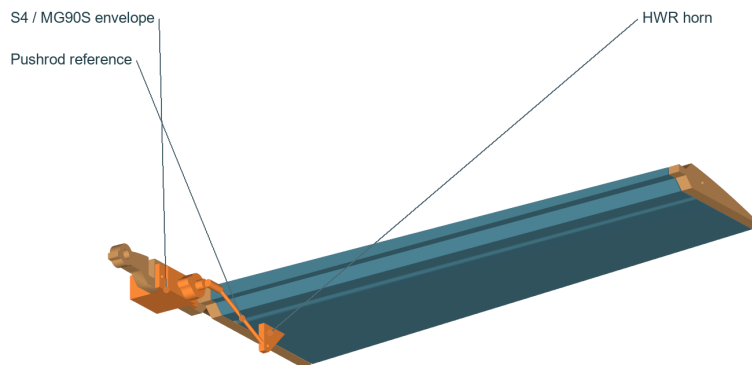
Use A21 on the right and A11 on the left. These are the existing unmodified print geometries supplied again because they were missed. Use their 2 mm common-axis pilots, not a full-length hinge shaft.

Measure the actual spoke/pin diameter. Align the two end pilots and finish them carefully to a smooth rotating fit. Each aileron uses two short end pins. Keep adhesive out of the bearing and keep the surface clear of its fixed rail.

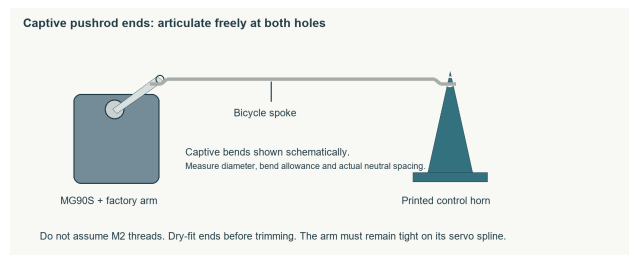
Make bent L heads from suitable spoke offcuts, and trap each head against the wooden endplate with reinforced tape or a tie. The head must remain clear of the moving surface and cannot slide out. Test the restraint in both directions; replace tape if it peels or allows pin movement.

Check before continuing: Both ailerons move freely; four end pins are positively captive without bearing glue or hinge rub.

Mount wing servos, horns and pushrods



Actual airframe, horns and source servo envelopes. Arm and pushrod cylinders illustrate a route; set length and captive bends on the real parts.



Fit an MG90S at each reinforced plywood outer-rib pocket: S3 left and S4 right. Before fitting arms, keep flight battery/ESC disconnected and prop removed. Connect one servo to a standalone tester: verified signal to signal, positive to regulated 5 V and ground to ground, following its keyed contact view. Select neutral mode; then fit the factory arm near a right angle to the pushrod and retain its supplied centre screw.

If no verified 5 V tester is available, leave arms off and skins open. Complete the bench power/configuration checks in Steps 26-29 and 30-31, command neutral from the configured FC, then return here and fit the arms. Do not close wings at Step 22 before this dependency and actual full-travel tests pass.

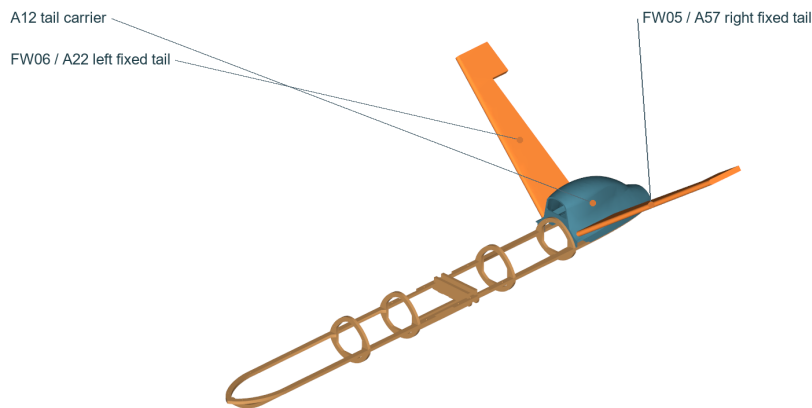
Seat the real mounting tabs against the actual rib. Use the servo supplied mounting hardware only after matching tab holes and screw/pilot dimensions; mark and hand-drill the omitted A44 mounting holes only after checking actual tab spacing and safe pilot/edge distances. The source servo envelope does not prove MG90S tab fit. If the tabs cannot be securely captured, hold this joint for measured fitting.

Fit HWL/HWR to their respective moving ailerons at the CAD contact. Abrade only the bond patch, clean and use the qualified adhesive through full cure. Horn pilot holes are nominal 2 mm for the pushrod, not servo mounting holes.

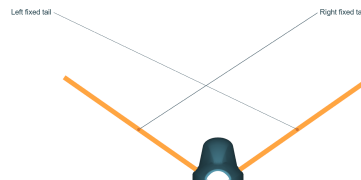
Trial one bicycle-spoke pushrod per wing with captive articulating ends. Measure its diameter, thread, arm hole and horn hole before cutting. The pictured 10 mm arm radius and straight rod are references, not a cut-length instruction. Leave all adjusters and the servo arm screw reachable.

Check before continuing: Servos and horns cannot shift under light control load; linkage moves without over-centre motion, rub or pin escape.

Form and fit the fixed upright V-tail



Upper: evolving body assembly. Lower: rear close-up shows both fixed tails seated symmetrically in the actual A12 carrier.



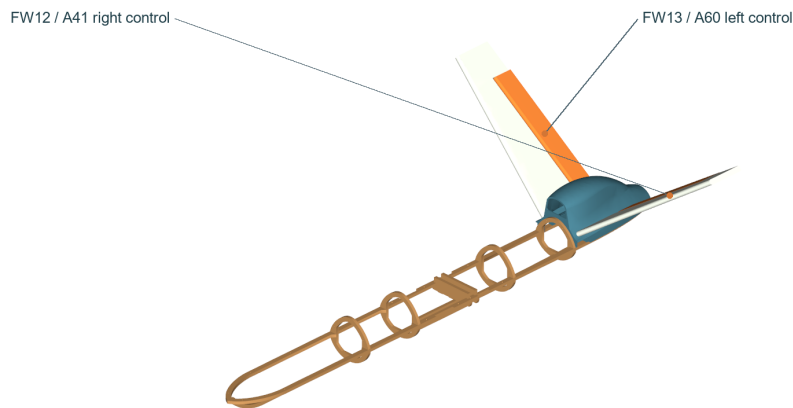
FW05 becomes the right fixed tail A57; FW06 becomes the left fixed tail A22. Each is one whole folded skin including its root collar. Match the source tail profile and trial the root against A12.

Precurve and fold progressively, keeping the outer paper intact. Remove or compress inner foam only where the fold and root need relief. Test on an offcut first. Dry-fit both fixed tails before bonding so their angles and trailing edges are symmetric.

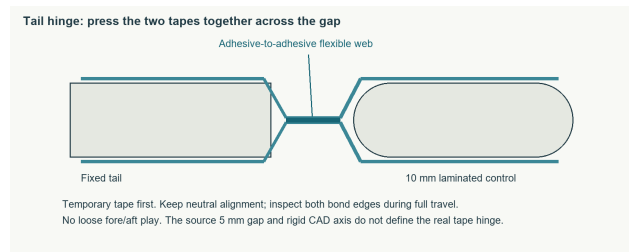
Bond the tail skin and its A12 seating patch with a tested foam-safe adhesive that also suits the printed carrier where they contact. Support both tails at the CAD angles through full cure; do not force a warped tail into a socket.

Check before continuing: Tail roots seat in A12, both halves angle upward symmetrically and no paper is torn or core crushed.

Laminate the moving tail controls and hinge them



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.



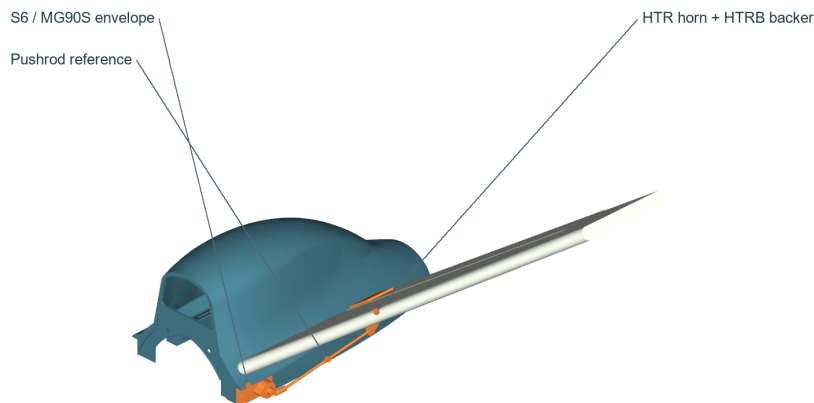
FW12 right uses two 5 mm blanks to make A41; FW13 left uses two to make A60. Bond matching layers flat with the qualified foam adhesive. Keep supported through full cure, then finish the marked outline, trim allowance and rounded leading edge.

Hold each moving surface at neutral next to the fixed trailing edge. Apply reinforced tape to both faces and press the exposed tape together across the gap to form a flexible web. Keep the gap free from a hard glue bridge.

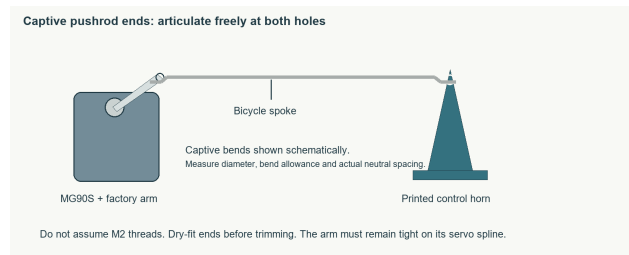
Trial movement roughly 15 degrees each way as a basic bench fit check, not a flight-qualified throw. The web must not peel, bind or allow fore-aft slop. Stop and remake a hinge that tears the paper or cannot move symmetrically.

Check before continuing: Both laminated controls are flat, retained by sound flexible webs and move freely in both directions.

Fit the two tail servos and backed horns



Right-side tail close-up; mirror the hardware for the left. Source servo envelope and schematic rod route require actual tab and linkage fitting.



S5 is the left tail MG90S and S6 the right. The source shaft targets are X-63.85/+63.85, Y411.10, Z6.215. They sit beside the original tail roots, with arms, centre screws and pushrod adjustments accessible.

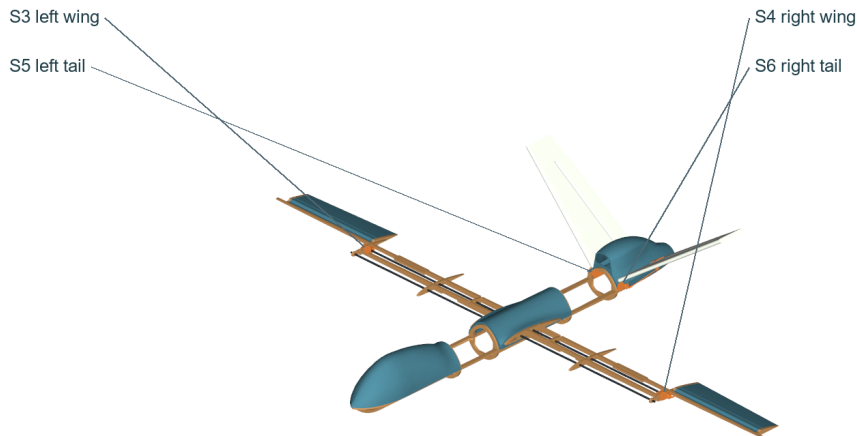
Use the verified 5 V tester/neutral procedure in Step 16 before fitting the factory arms and retaining the arm screws. With no tester, leave arms off and return after the FC power/configuration checks; keep the skins open. Trial actual body/tabs, and use matching supplied hardware at sound mounting contact only. Hold a joint if the envelope does not securely seat the real unit.

Fit HTL/HTR horns to the left/right moving controls with HTLB/HTRB backing plates against the opposite foam face. The backers spread local load. Fit and cure the tested foam/print bond without crushing the core. Do not trust a horn glued to unsupported paper.

Use two more bicycle spokes as tail pushrods with measured, captive articulating ends. Cut only after full neutral and combined pitch/yaw movement has been tested. Keep rods clear of the carrier and motor wiring.

Check before continuing: Both tail servo mounts, horns and backers remain secure; every combined endpoint moves without rubbing or foam peel.

Check the complete open structure



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

Put the body, both wings, fixed tails and controls together before any skin is glued closed. Confirm all wood laps are cured, printed root contacts are sound and rods and hinge pins are retained.

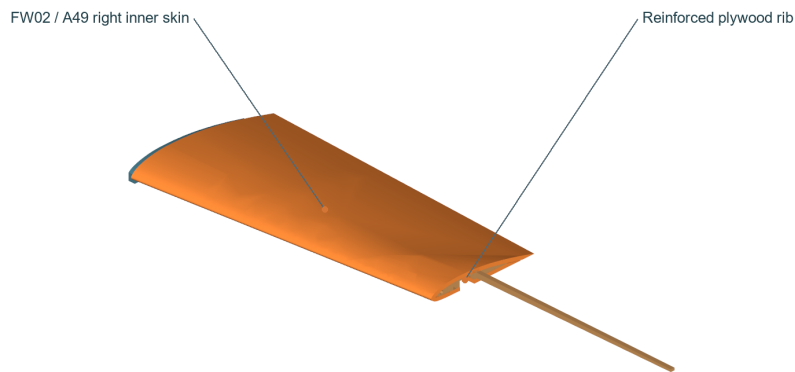
Move each control through full intended bench travel while supporting the airframe. Check rubbing, play, horn peel, servo buzz and an arm approaching over-centre. Correct the mechanical cause before limiting travel in software.

Run each wing servo cable to its root with enough slack for wing handling. Use smooth-edge tape/tie supports on fixed structure, avoiding hinges, pushrods and rod movement. Plan removable wing connectors and label S3 left / S4 right at both ends. Run tail cables forward along the frame, clear of the motor and driver path.

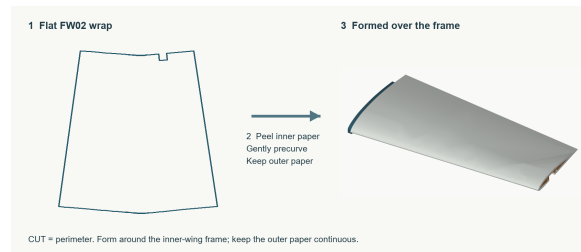
Leave an accessible route to the servo mounting and arm screws, connector separation points and tail adjusters. If you cannot service one of them after skinning, change the access plan now.

Check before continuing: Four controls move freely, every connector is labelled, and all restraints and service points can be inspected.

Form the inner wing skins you already cut



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.



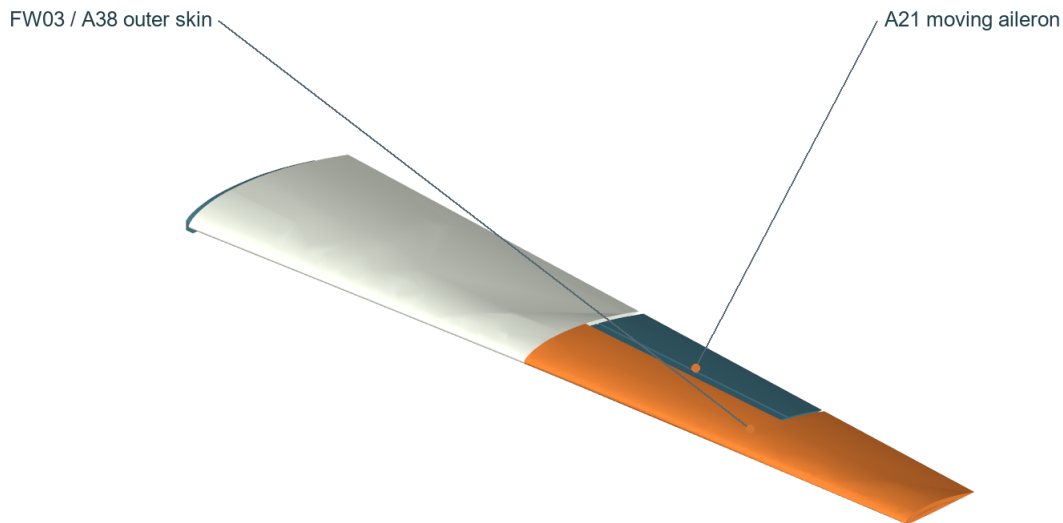
FW02 is the right inner skin (A49) and FW01 the left (A52). If each is already a whole skin, keep it intact. If supplied as ROOT/TIP cuts, butt the matching straight edges on a flat board, bond the core with foam-safe glue and tape across the outer paper. Cure the join before curving.

Mark the inside, leading edge and part ID from the cut map. Curve progressively over the actual inner wing frame; keep the outer paper continuous. Only relieve the inner core where a rib, rod or fold interferes. The source fold marks are hand references, not extra through-cuts.

Dry-fit the root, inner/middle ribs and rods together. Hold the airfoil without twisting it. Do not hide a broken spar or force a poor root fit under the skin.

Check before continuing: Each skin matches its side, forms without buckles and clears the whole rods and root without forcing.

Dry-fit outer skins and keep access open



Actual outer skin shown in its final position, held temporarily for fit. Permanent closure waits for powered controls and supply checks.

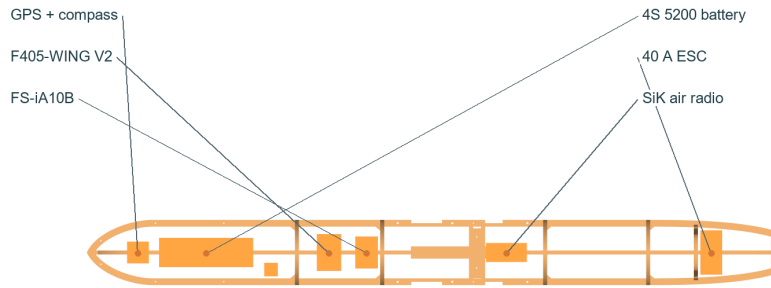
FW03 is the right outer skin A38 and FW04 the left outer skin A27. Trial around the completed ribs, fixed rails, servos and controls. Retain the documented inner-foam relief against A10_R/A53_R and their left mates. Relieve upper and lower inner foam locally near the closed front rod seat on both wings; preserve outer paper. Rods must already be inserted axially before closure.

Confirm rod/root retention, captive aileron pins, horn bond and servo mounting. Route the servo cable, test its connector and leave the servo arm accessible. Keep an inspectable/removable seam or opening at each servo rather than permanently burying it.

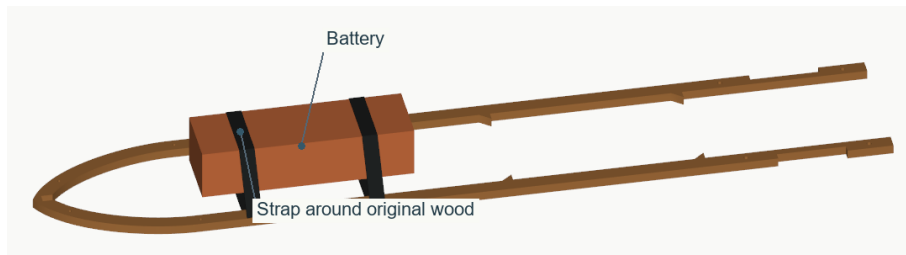
Hold the skins with removable tape now. Leave them unbonded until FC configuration, actual Vx/Vx2 loaded-servo movement, direction and failsafe tests pass. Final wing bonding is in Step 34; keep tape clear of the moving hinges.

Check before continuing: Both wings dry-fit symmetrically with smooth cable exits; all joints, servos and connectors remain accessible.

Place and restrain the battery and electronics



Top view of actual frame with nominal device envelopes. Final positions follow real connector access, interference tests and measured balance.



Start with the battery on sound A13 wood in the forward bay; the old centre target Y-260 is only a trial location. Use an insulating nonslip pad and two independent straps around the existing frame and battery. Round or pad sharp contact edges. Adhesive may locate the pad but must not be the only pack restraint.

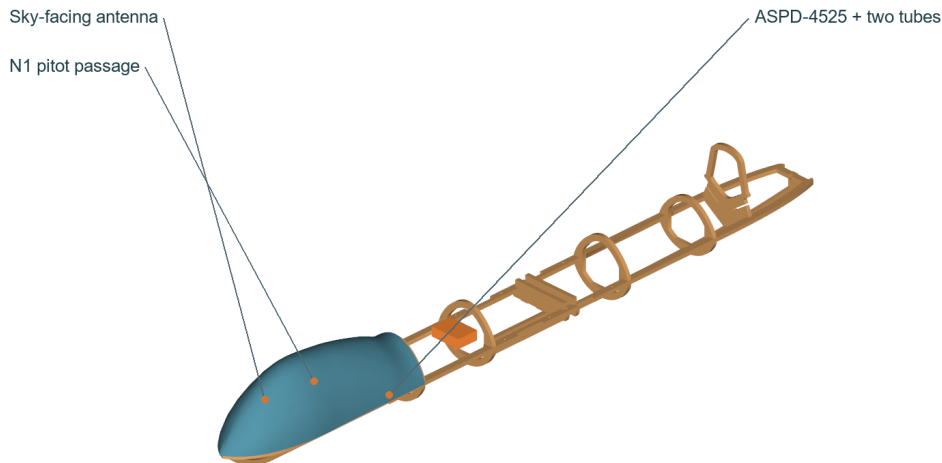
Fit F405-WING V2 with known arrow/orientation on a suitable insulated vibration pad near the forward centre body. Leave USB, microSD and all plugs reachable. Place FS-iA10B nearby with separated antenna tips away from carbon and motor-current wiring.

Place the SiK AIR unit in the middle/aft body where its antenna and serial plug remain serviceable. Mount ReadytoSky 40A ESC aft on sound frame with ties/tape, cooling surface exposed and motor wires secured. Check its lead-length guidance before adding battery-wire length.

These drawn device boxes are nominal envelopes placed on real CAD, not exact device models or a fit certification. Trial the real leads, straps, cover and removal path. The old analysis already shows battery positions near Y-192 can collide with A34; do not force that position to chase a predicted CG.

Check before continuing: Battery cannot slip and can be removed unplugged; board USB/SD, receiver plug, radio and ESC remain accessible.

Place GPS, pitot and airspeed tubes



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

Place the documented M9N-5883 GPS/compass with antenna face skyward in the upper nose region. Keep the compass at least 100 mm from power wiring, ESC, motor and ferrous hardware as the manufacturer recommends; centre-to-centre box spacing alone is insufficient. Verify the real clearance to leads.

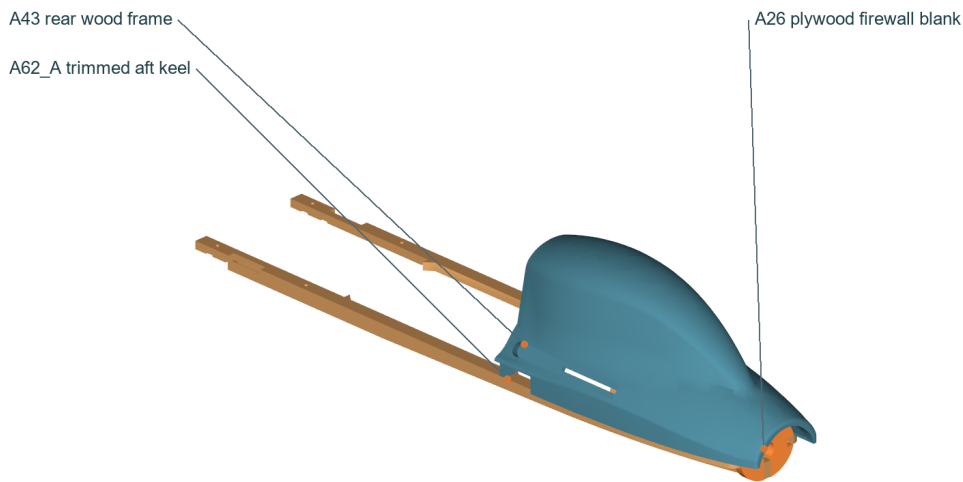
Reuse the owned A59/N1 nose. With electronics and battery removed, offer up the actual pitot and mark a forward-facing passage that clears the shell, frame and nose removal path. Drill a controlled pilot and enlarge only for a snug tube/support fit. Do not copy a guessed CAD hole location.

Secure the pitot to sound support with tape/ties without covering sensing ports. Lead both silicone tubes to ASPD-4525 near the forward body; support gently, avoid kinks and leave the sensor ports reachable. Identify total/static ports from the actual sensor/manual.

The pressure sensor, GPS and radio must match their documented models before connector pin order is used. If a label differs, stop that connection and identify its exact manual; continue the other mechanical work.

Check before continuing: Nose removal does not pull the tubes; sensing holes remain open and compass separation is measured.

Dry-fit the plywood motor firewall blank



Actual no-motor-hole plywood A26 in the rear frame. The nominal digital insertion route is clear; real stock fit and cured joint strength still need physical proof.

Use the current nominal 6 mm plywood A26 no-motor-hole blank. It retains the source shaft/keyhole opening and smooth 6 mm-height flank seats. No motor screw pattern is laser cut; the actual EMAX base must set the drill pattern.

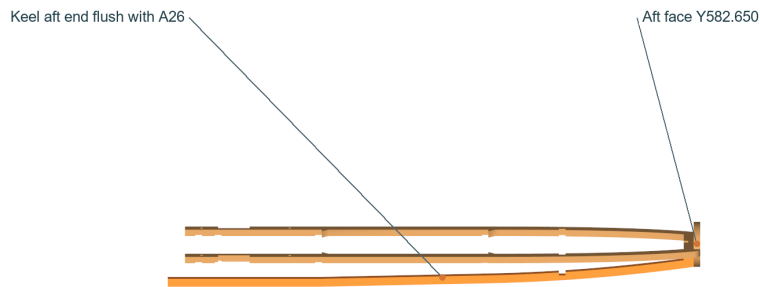
Insert the mount axially from the rear with the motor removed. A43 blocks a drop-in vertical approach. Keep the aft keel flush with the firewall aft face at Y582.650; the current A62_A cut outline already removes the former 13.015 mm rear projection.

Trial A26 on actual A43/A62_A and check flatness, slot fit, A12 clearance and driver access. The central slot and keel are both nominally 6 mm, so actual stock, kerf and adhesive determine fit. A43 does not fully fill the flank notches; collision-free CAD does not prove a complete tongue joint.

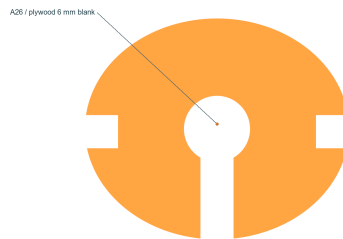
Physical mount mismatch has not been confirmed. If a seat catches, identify the exact contact and measure it before local fitting. Test the actual wood adhesive/contact on an offcut, bond only after the dry-fit and hold square through full cure. Keep the shell open for real motor checks.

Check before continuing: Mount seats by the rear axial route, actual contact and driver access are identified, and no unmeasured motor holes are drilled.

Motor fitting detail and manual drilling



Current trimmed keel and plywood firewall blank. Lower: the source shaft/keyhole relief remains; motor bolt holes are intentionally absent.



Check the documented EMAX ECO II 2807 1300KV label on the real motor. Its official drawing specifies four M3 threads on a 19 mm pitch-circle diameter, equivalent to 13.435 mm adjacent square spacing. This is a reference, not permission to drill a guessed pattern.

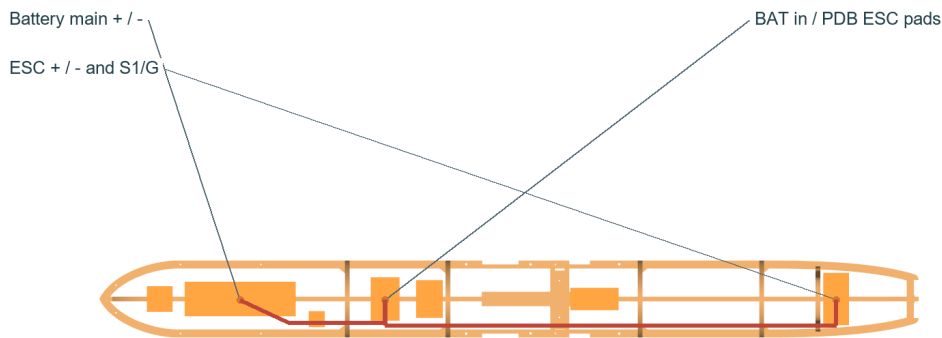
Offer the actual motor base or its verified template to the blank at shaft centre X approximately 0, Z-2.5. Confirm shaft/boss relief, bell/wire clearance, flange seating and full driver access, then mark and hand-drill the matching clearance holes. Keep the propeller removed.

Measure the full screw/washer/6 mm plywood stack, head diameter and safe M3 thread depth. Supplied M3x8/M3x10 lengths are not automatically correct. Screws must clamp without bottoming or reaching windings; any washer must sit fully supported and clear the bell.

The aft keel ends flush at Y582.650 in the current cut set. For an older blank ending at Y595.665, remove only its 13.015 mm aft projection, keeping the joint and all wood forward of the firewall. Qualify actual mount contact, screw retention, thrust/torque, heat and vibration progressively before flight.

Check before continuing: Real motor pattern, boss seating, safe screw engagement and driver approach pass before fastening or powered testing.

Connect high-current power with the prop off



Red traces show power routing on the real frame. Match the actual wire lengths and service loops; device boxes are nominal envelopes.

Keep the GenX RKI-4897 4S 5200 mAh pack disconnected. Its nominal voltage is 14.8 V and full voltage 16.8 V. Check the actual matching high-current main connector, positive and negative; the balance plug is for the charger, not the FC.

Connect the battery main positive/negative to F405-WING V2 BAT input with the verified connector and suitable rated wiring. Connect ESC thick positive/negative to the board PDB ESC power pads, following their actual labels. Connect the three motor phase wires to the ESC and restrain them clear of the bell and controls.

Connect ESC throttle signal to S1 and its control ground to G. The ReadytoSky family has a documented BEC despite some OPTO titles: inspect and meter your unit. If its red BEC wire is live, isolate and insulate that positive so it is not accidentally paralleled with onboard servo power.

Insulate every joint and support wires at frame ties without crushing insulation. With the battery still unplugged, check continuity, shorts, polarity and connector key orientation with a multimeter. Route high-current wiring away from compass, receiver antennas and sensor cables.

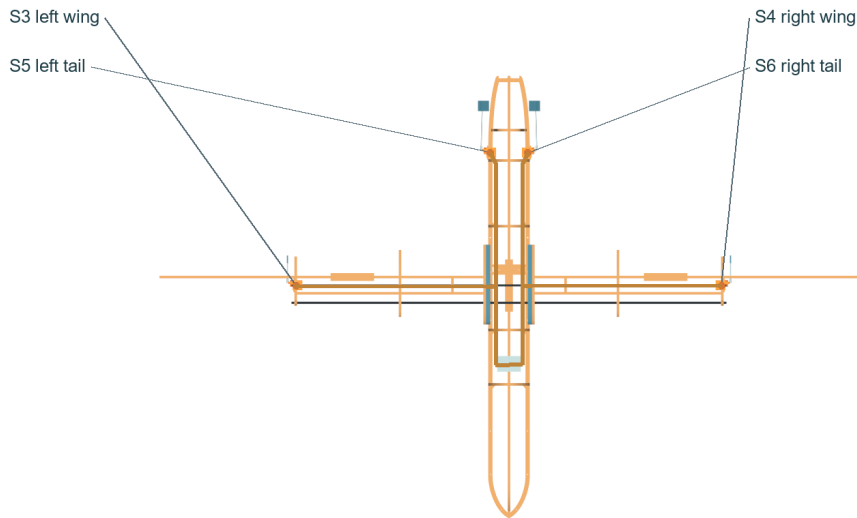
Check before continuing: Polarity and insulation pass; ESC BEC handling is verified; no motor wire can reach a moving part.

Power and motor connection map

From	To / net	Check
Battery main + / -	FC BAT + / GND	4S 14.8 V nominal, 16.8 V full; connector polarity
FC/PDB ESC + / -	ESC DC + / -	Wire and pad rating; continuity; insulated joints
ESC three phase leads	Motor three phase leads	Prop off; secure away from rotating bell
FC S1 / G	ESC throttle signal / ground	Live ESC BEC positive isolated if onboard BEC used
FC Vx set to 5 V	S3 and S4 servo + rail	Measure before plugging in servos
FC Vx bridge to Vx2	S5 / S6 servo + rail; S7 later	Vx2 is unpowered without bridge or separate supply
FC S3 / S4 / S5 / S6	Each servo signal	Each servo also has +5 V and common GND

Check before continuing: Use actual board pad marks and actual keyed contacts; wire colour alone does not establish polarity.

Connect servo power and receiver



Amber traces show fixed-frame harness routes from the four source servo locations to the FC. Leave labelled detachable connectors at wing roots. S7 is deferred.

Set the F405-WING V2 servo BEC Vx to 5 V for MG90S. Bridge Vx to Vx2 when using the onboard supply for the tail servos. Vx powers S3/S4; Vx2 powers S5/S6 and later S7. Measure both rails before connecting servo plugs.

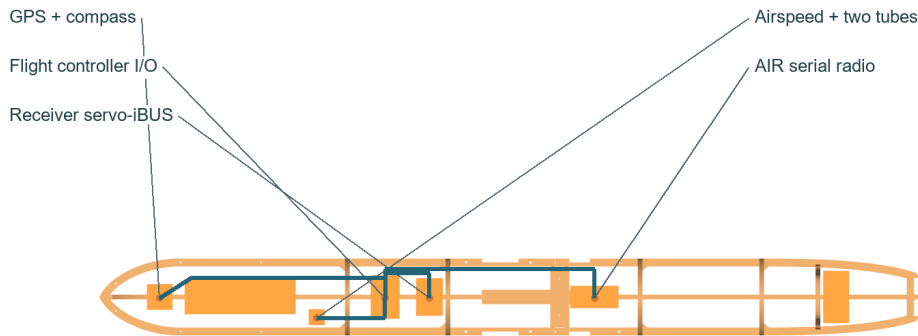
Connect each servo lead signal, positive and ground to its matching output: S3 left aileron, S4 right aileron, S5 left V-tail, S6 right V-tail. Confirm the actual plug orientation. Keep S7 steering unplugged until gear is fitted after midsems.

Connect FS-iA10B servo-iBUS output signal to FC RX2, plus rated peripheral 5 V and common GND. Do not use the receiver SENS port. Label the wing connectors on both ends; protect their latch/release access.

Onboard Vx is rated 5 A continuous and 6 A peak; peripheral 5 V is rated 2 A total. Test actual simultaneous servo and accessory load before closing. Add up rated accessory consumption rather than assuming every 5 V connector has a separate supply.

Check before continuing: Each plug polarity and side is checked; Vx and Vx2 read 5 V; receiver servo-iBUS is on RX2.

Connect sensors and the SiK air radio



Teal traces show sensor/receiver/radio routes on the real frame. Keep signal routes clear of high-current wiring; final plug orientation comes from the actual devices.

For documented M9N-5883, connect GPS TX to FC RX3 and GPS RX to FC TX3, plus rated 5 V/GND. Connect compass SDA to DA2 and SCL to CL2. TX crosses to RX; do not join TX to TX.

For documented ASPD-4525, connect its SDA to DA2 and SCL to CL2 on the same I2C2 bus, plus 5 V/GND. Its manufacturer GH order is 5V, SCL, SDA, GND. Check connector view/key on the actual device before fitting it; do not infer numbering from the aircraft picture.

Connect RKI-1980 SiK AIR TX to FC RX1 and AIR RX to FC TX1, with verified 5 V and GND. A standard 3DR GH-6 harness uses 1=5V, 2=FC TX1, 3=FC RX1, 4/5=CTS/RTS unused here, 6=GND; verify key orientation and continuity of your owned harness.

The black USB GROUND radio plugs into the computer, never into aircraft UART pads. Keep signal grounds common. Use a verified splitter/parallel harness for shared I2C; insulate unused contacts and support solder joints.

Check before continuing: Every labelled net reaches its intended device, no supply/signal contact is swapped, and I2C harness continuity passes.

Receiver, sensor and radio map

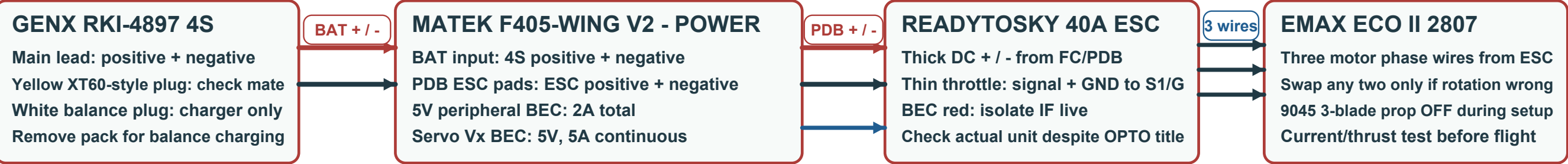
Device pin / net	F405-WING V2 pad	Notes
FS-IA10B servo-IBUS signal	RX2	BRD_ALT_CONFIG=0; not SENS
Receiver + / GND	Peripheral 5V / GND	Check actual keyed harness
M9N GPS TX / RX	RX3 / TX3	Cross serial directions
M9N compass SDA / SCL	DA2 / CL2	I2C2 shared with airspeed
ASPD-4525 SDA / SCL	DA2 / CL2	GH order 5V, SCL, SDA, GND
GPS and ASPD power / GND	Peripheral 5V / GND	2 A total peripheral BEC budget
SiK AIR TX / RX	RX1 / TX1	Cross serial; verify real harness
SiK AIR power / GND	Verified 5V / GND	GROUND USB unit stays at computer
Pitot total / static pressure	ASPD total / static ports	Two tubes; identify real ports; no kinks

Check before continuing: If any received device differs from the named model, identify its manual before connecting it.

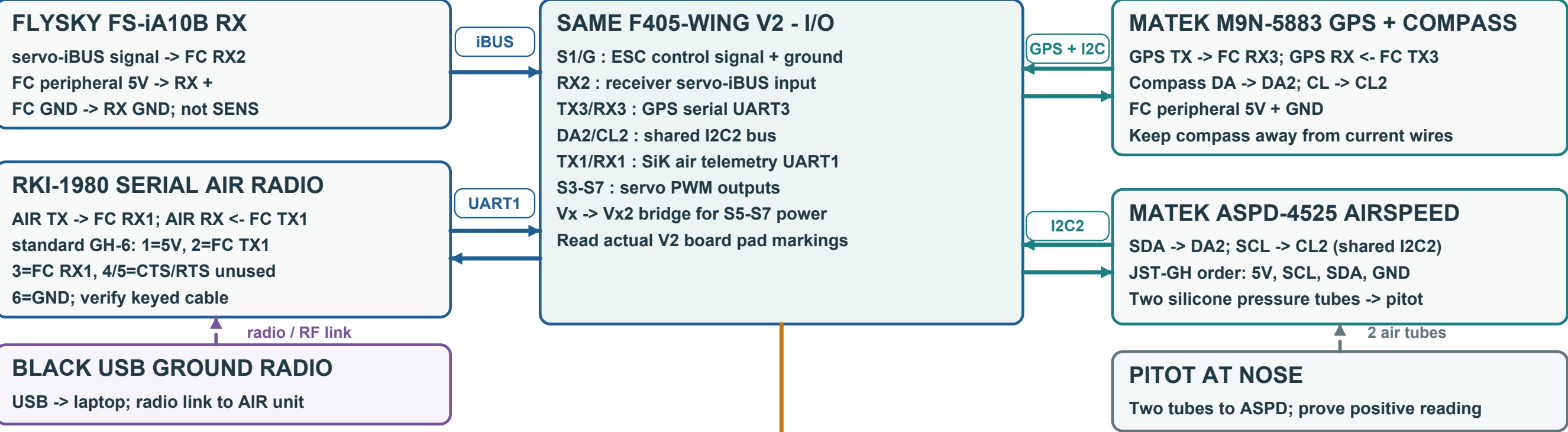
REAPER - COMPLETE ELECTRONICS CONNECTION MAP

One diagram for the owned parts. This is a functional net map, not a physical connector-position drawing.

1 HIGH-CURRENT POWER + MOTOR

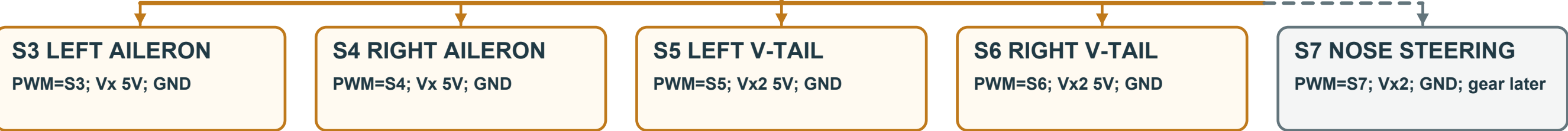


2 FLIGHT-CONTROLLER SIGNALS + RATED 5V PERIPHERALS



3 FIVE MG90S SERVOS - EACH LEAD HAS SIGNAL, +5V AND GND

Use Vx 5V for S3/S4. Bridge Vx to Vx2 for S5/S6/S7 when using the onboard servo BEC.



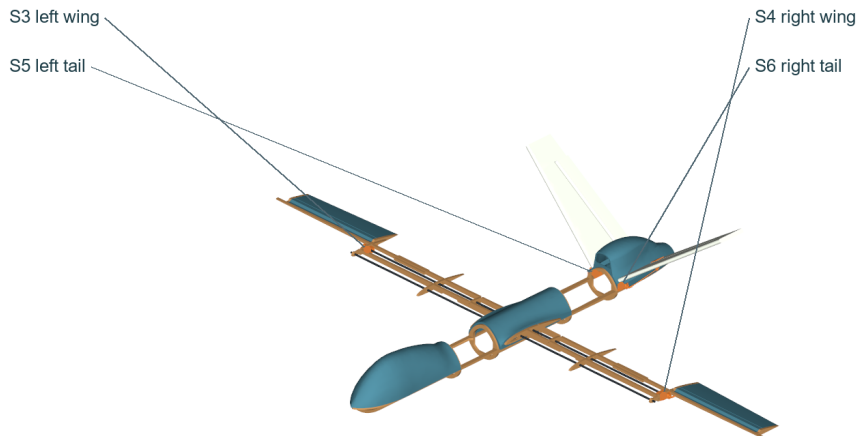
BEFORE POWER: Propeller off; verify polarity and plug/key order. Isolate ESC BEC red only if live.

BEFORE FLIGHT: Prove loaded Vx/Vx2, controls/failsafe, pitot, motor current/thrust and CG.

Solid lines = electrical route; purple dashed = radio link; gray dashed = air tubes/optional S7.

Functional connections - check actual keyed contacts

Power carefully and prove the open installation



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

Propeller removed, transmitter on with throttle low: first power the isolated board by USB for inspection. After continuity/polarity checks, test the flight-power wiring with an appropriate current-limited source. Do not exceed any input range.

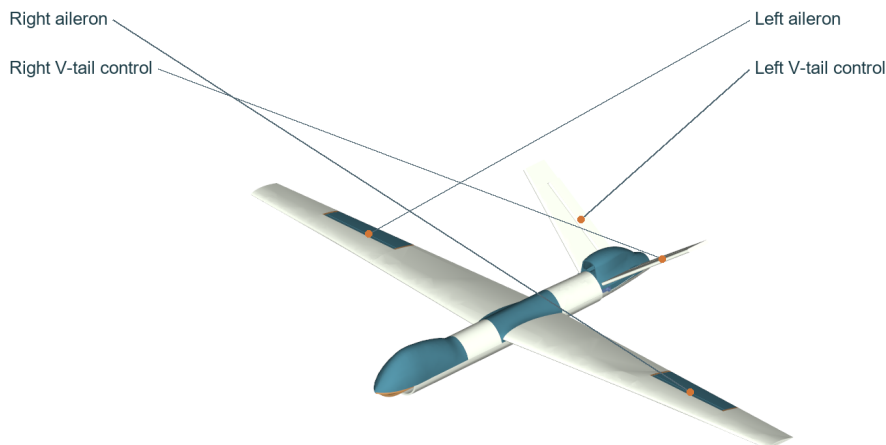
Measure V_x/V_{x2} and peripheral 5 V. Connect accessories one group at a time, watching for heat, wrong polarity and unusual current. Leave servo movement for the following firmware/output checks; then test all four simultaneously on the actual V_x/V_{x2} rails under a representative light control load before final skin closure.

Save the observed voltages, load current and any fault. Disconnect battery power before swapping a plug. Correct a mechanical bind before increasing power or adjusting endpoints.

Check nose and cover removal with all harnesses in place. Mark every detachable connector and leave a short service loop without a loose coil near the motor. Take clear photos of the complete open installation for later service.

Check before continuing: Polarity and idle rails pass; no heat or resets; powered servo-load proof stays open until after output configuration.

Install firmware and save the starting settings



Aircraft orientation reference for checking the ground-station attitude display; software values require readback on the real controller.

With the propeller removed, connect the F405-WING V2 to Mission Planner over USB. In Initial Setup > Install Firmware choose Plane and the exact MatekF405-Wing target supported for the V2 board. Follow the current board instructions; do not flash a similarly named target by appearance.

Connect, record firmware version and board identity, then save a parameter file in Config/Tuning > Full Parameter List before changes. In the attitude display move the airframe by hand: nose down/up and right/left roll must match the display. Set board/compass orientation to the real installation.

Set BRD_ALT_CONFIG=0 for iBUS on RX2 and check channel bars in Initial Setup > Radio Calibration. Turn transmitter V-tail mixing off so the controller owns the mix. Reboot when a parameter instruction requires it, then read the value back.

Menu wording can vary by ground-station version. This procedure is an instruction for your hardware; no aircraft was connected or firmware flashed while creating this guide.

Check before continuing: Correct target/version and baseline file recorded; attitude and every receiver channel respond in the correct sense.

Assign the actual outputs explicitly

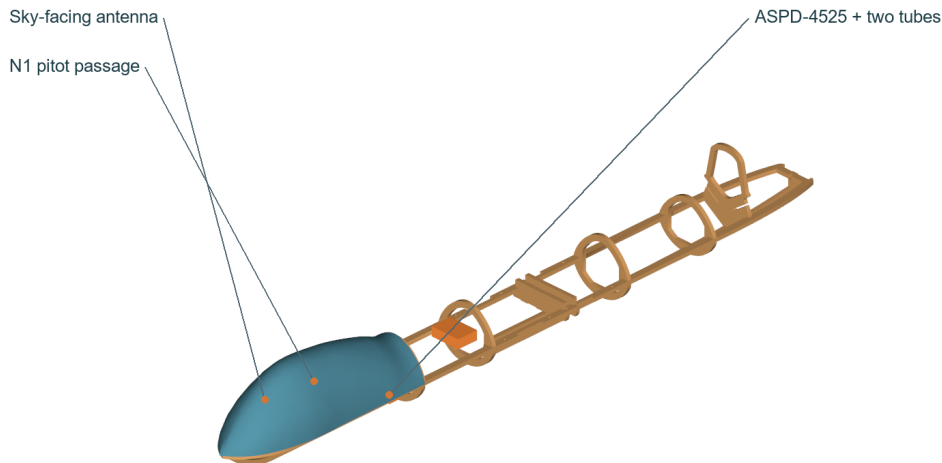
In Full Parameter List set and write the output functions below. Read them back after restart. Output IDs do not establish correct movement; Step 33 proves the directions.

Do not import a ready-to-fly file. Keep S7 steering deferred until measured gear fitting. Start with conservative neutral/endpoints established from free mechanical travel; flight throws require experienced pilot acceptance.

Parameter	Starting value	Actual device
SERVO1_FUNCTION	70	Throttle / ESC S1
SERVO3_FUNCTION	4	Left aileron S3
SERVO4_FUNCTION	4	Right aileron S4
SERVO5_FUNCTION	79	Left V-tail S5
SERVO6_FUNCTION	80	Right V-tail S6
SERVO7_FUNCTION	26 later	GroundSteering only after gear fitted
BRD_ALT_CONFIG	0	iBUS RX2
SERIAL3_PROTOCOL	5 if required	GPS on UART3
BATT_VOLT_MULT / BATT_AMP_PERVLT	11.0 / 66.7 for V2	Calibrate readings against meter
ARSPD_TYPE / ARSPD_BUS	1 / 1 for ASPD-4525	Verify actual firmware bus mapping

Check before continuing: Write, restart as required and read back; final reversal, endpoints and failsafe still need physical tests.

Calibrate radio, attitude, compass and airspeed



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

Propeller off: follow Radio Calibration and Accelerometer Calibration in Initial Setup. Set known board and compass orientations. Recheck displayed roll/pitch and save changed parameters. Check SiK UART1 and the air/ground pair settings before claiming telemetry works.

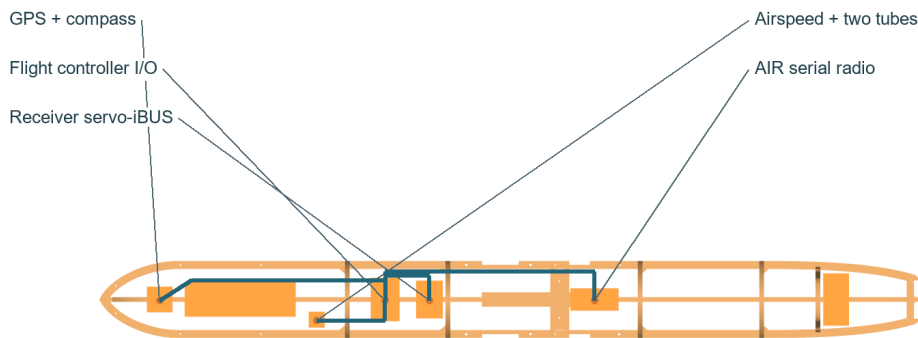
Take the aircraft outdoors away from metal and power interference for compass calibration and GPS lock. Check live satellite/position data; keep the compass at its final mounted location. Recheck interference after the power harness is fixed.

Zero ASPD-4525 in still air and inspect both pressure tubes. Use the current ArduPilot airspeed setup for the actual firmware; ARSPD_TUBE_ORDER=2 is auto only if supported. Gently direct airflow toward the pitot and confirm a positive reading. Do not blow strongly into the sensor or assume swapped ports are acceptable.

Compare battery voltage/current with a suitable measured reference. The V2 starting scales are 11.0 and 66.7 A/V, not the older V1 current scale. A displayed sensible number alone is not a calibration.

Check before continuing: Radio endpoints, attitude signs, compass orientation, GNSS and positive airspeed response are recorded; voltage/current agree with measurement.

Pair the SiK radios and prove telemetry



The AIR unit in the frame sends telemetry; the GROUND USB unit remains at the computer. Teal lines show actual-frame harness routes.

Fit the correct antennas to both radios before powering them. With the ground station disconnected from the vehicle, choose the GROUND USB radio COM port at 57600 baud. In Mission Planner open Initial Setup > Optional Hardware > SiK Radio and Load Settings for local and remote units.

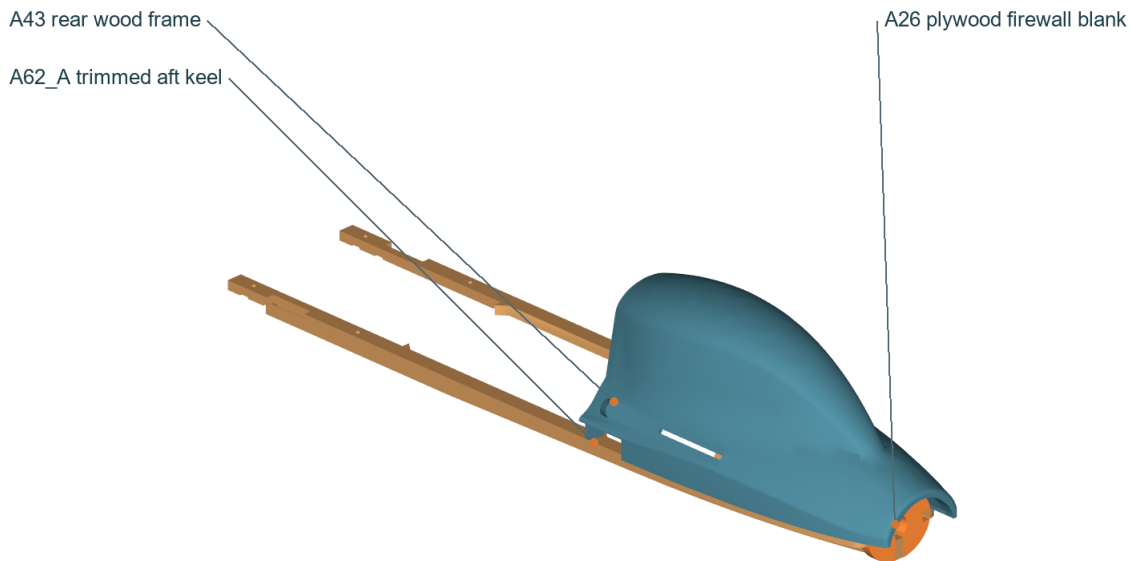
Inspect the pair before changing good defaults. Net ID and required Air Speed/ECC settings must agree. If they do not, use Copy Required Items to Remote and Save Settings per the current SiK procedure. AIR serial baud must match FC UART1; GROUND serial baud must match the computer. Do not invent radio transmit power or range.

On the F405 set SERIAL1_PROTOCOL=1 for MAVLink1, or 2 only if the radio firmware supports MAVLink2; SERIAL1_BAUD=57 means 57600. Write, restart as required and read back the values. These settings apply to TX1/RX1, not the GPS UART.

A solid green SiK link LED indicates a paired link; blinking indicates searching. Select the GROUND COM at its matching baud and Connect. Check live heartbeat/parameters and tilt the aircraft to see Flight Data attitude update over radio. If it fails, disconnect flight power before checking crossed TX/RX, supply/ground, baud and pair settings. Keep USB available for recovery.

Check before continuing: Actual radio-link connection and live attitude/parameters pass; a lit power LED alone does not prove telemetry.

Check motor direction without a propeller



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

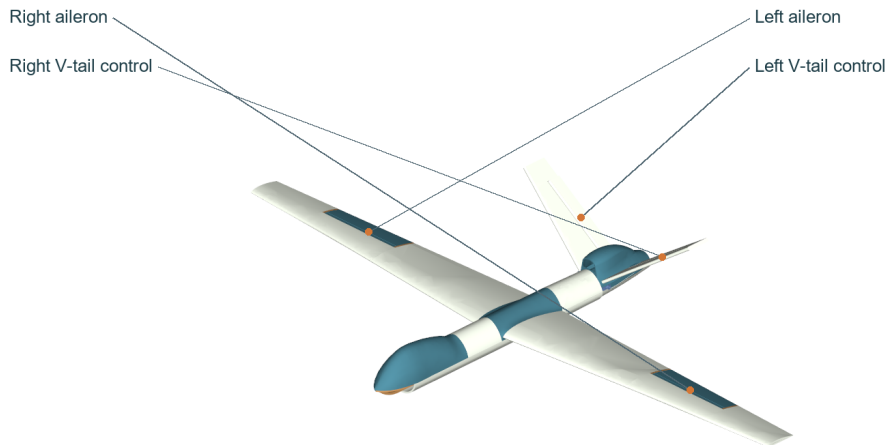
Secure the airframe, clear the motor bell and keep the propeller removed. Follow the actual ESC calibration procedure if its protocol requires it; do not run a generic calibration blindly.

Command only a short low-power direction test with a means to disconnect power. Confirm the rear pusher rotates in the intended direction for the actual prop. If wrong, disconnect battery and swap any two motor phase leads; repeat the test.

Stop for vibration, rubbing, heat or loose mount contact. Check every motor screw and the frame joint afterward. This unloaded motor test does not qualify the 9045 three-blade prop on 4S.

Check before continuing: Rotation agrees with the actual pusher prop orientation; motor runs smoothly and the mount stays secure.

Prove every control direction and failsafe



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

In manual mode command roll right: right aileron trailing edge rises, left lowers; roll left reverses both. On this upright V-tail, pitch up raises both trailing edges, pitch down lowers both. Viewed from behind toward the nose, yaw right shifts both trailing edges toward aircraft right; yaw left shifts both toward aircraft left. Test pitch/yaw together and have an experienced builder verify the mix.

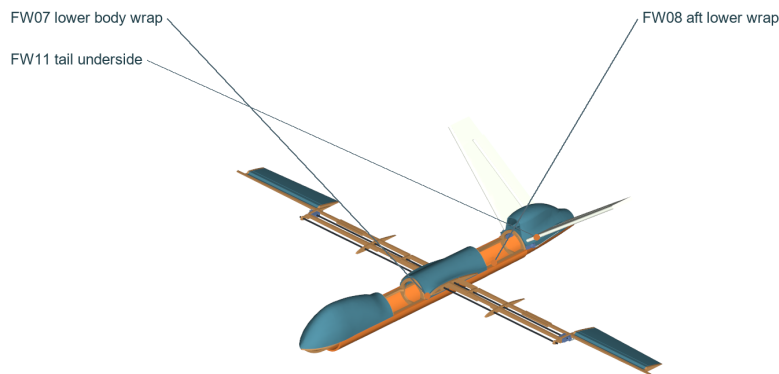
In stabilized mode raise the nose: both tail trailing edges must lower; lower the nose: both rise. Roll the right wing down: right aileron must lower and left rise to oppose the disturbance. If wrong, check board orientation, function assignment and the specific output reverse. Change one cause, then repeat every direction and mode.

At all combined endpoints check no bind, horn peel, pin escape, cable snag or servo stall. Record neutral, physical throws and endpoints. Reducing software travel must not hide a broken mount.

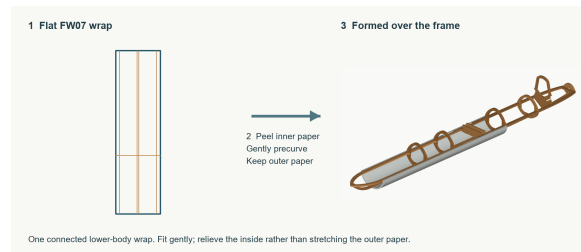
Propeller off, test transmitter loss in a controlled setting. Verify the intended safe throttle and mode action and its recovery on the actual receiver/controller, then perform the radio manufacturer range check. Repeat these tests after firmware, output, reversal or linkage changes.

Check before continuing: Manual and stabilized responses are correct in all directions; failsafe/recovery and free combined travel pass and are logged.

Fit lower body wraps after open checks



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.



First bond the previously dry-fitted wing skins after powered direction, failsafe and loaded Vx/Vx2 checks have passed. Use qualified foam adhesive progressively with each wing supported straight, retain servo service access, keep hinges clear and support through full cure. Repeat travel and symmetry checks.

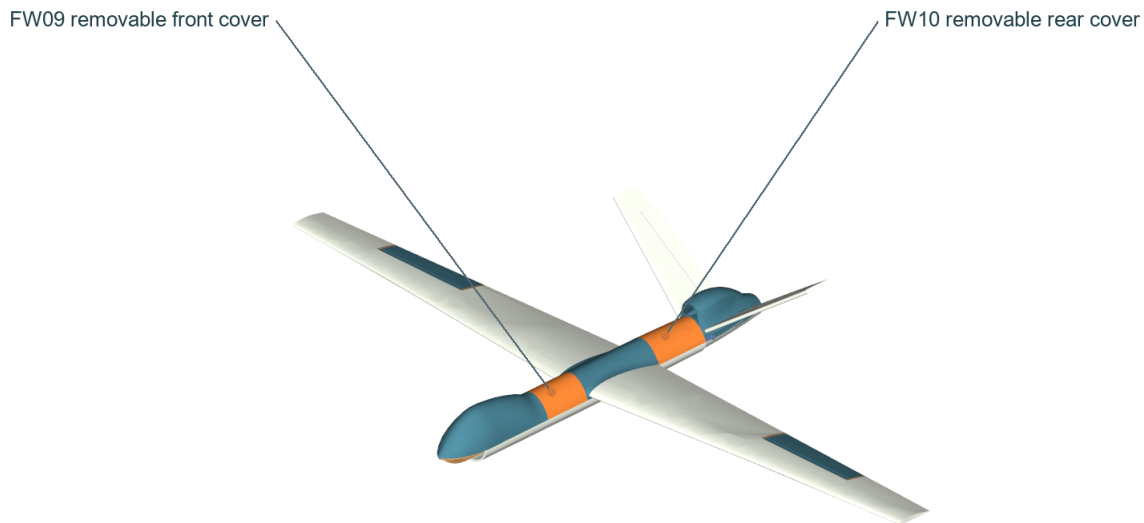
FW07 is the long lower body wrap replacing A56/A17 surface faces. FW08 covers the aft lower body A42. FW11 is the tail underside A20. Use the already-cut outlines; the FW07 205 mm cross-line is a hand station reference.

Precurve progressively, peeling inner paper where the documented wrap requires it while preserving outer paper. Trial FW11 in paper first; its compound curve may need short overlapping perimeter darts while remaining one connected piece. Do not flatten a buckle by crushing the core.

Keep the later gear openings and battery restraint path open. Confirm motor screw access, tail adjusters and wire routing once more, then bond fixed lower-wrap contacts progressively with the qualified foam adhesive. Hold shape through full cure and tape fixed seams.

Check before continuing: Lower wraps follow the frame, no wires are trapped, later gear access remains open and the nose remains removable.

Fit removable top covers and centre fairing



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

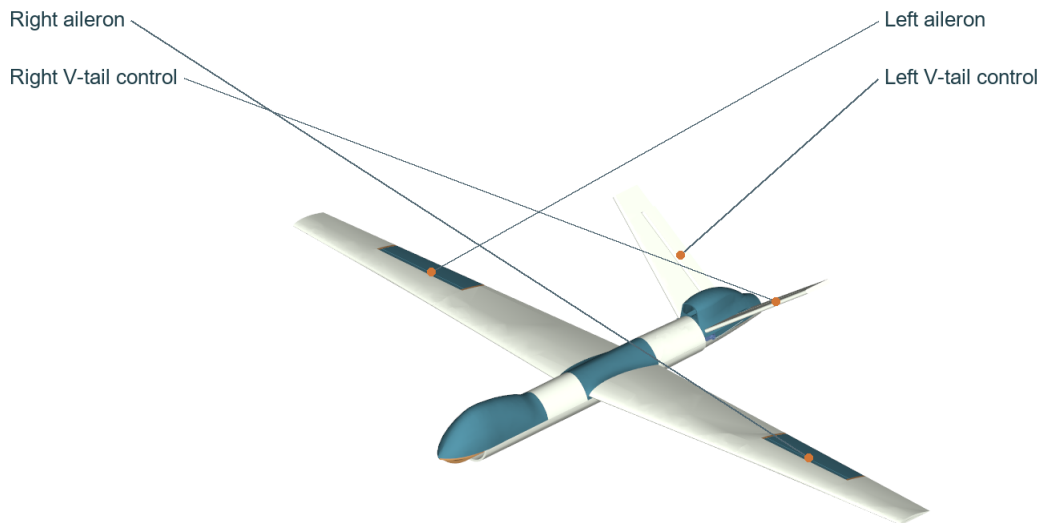
FW09 is the front upper body A08 and FW10 the aft upper body A15. Trial over the fully wired aircraft alongside A06 centre fairing, A59 nose and A12 tail carrier. Make only local dry-fit relief; do not recut a whole sheet.

Retain front/rear access covers with removable tape/ties at fixed skin/frame contact. Keep release edges reachable. Do not permanently glue closed the battery, FC USB/SD, wing connectors, tail linkage or motor service path.

Open and close each cover with the pack disconnected; remove and reinstall the battery without pulling a sensor hose or harness. Tape exposed fixed paper edges but keep all hinges free. Take the final access photos.

Check before continuing: Every required service point is reachable after closing; the pack is independently restrained and can be removed safely.

Park safely until gear work after midsems



Orange identifies parts added or checked now. All airframe shapes are placed CAD geometry.

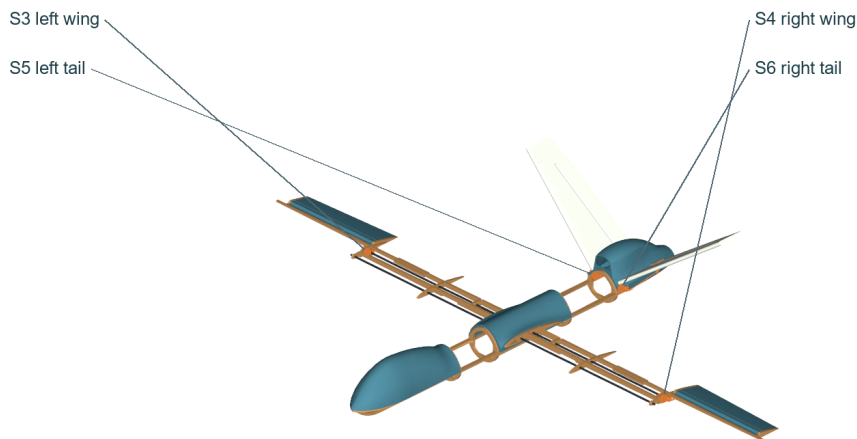
Landing gear fitting is deferred. Leave S7 and its steering linkage inactive, keep gear openings and A25/A47/A48 structure inspectable, and support the aircraft on padded cradles away from control surfaces.

After midsems measure the purchased gear, axle height, mounting footprint, steering clearances and supplied fasteners. Spread its landing loads into the wood skeleton; foam is not the landing-load mount. The original gear prints are not an automatic substitute.

Fit MG90S S7 only after its actual gear linkage is established. Set `SERVO7_FUNCTION=26` and appropriate `GROUND_STEER_ALT` per the current manual, then test direction and travel without binding. Obtain physical mount/load acceptance before taxiing.

Check before continuing: Deferred gear work is clearly identified; no taxi or flight is attempted with an unqualified attachment.

Physical acceptance gate 1: structure and balance



Inspect the internal load paths shown before and after each physical test; reopen service covers if needed.

Record the completed mass with the flight pack, straps, propeller/hub, final gear, wiring and fasteners. Measure lateral balance and fore/aft CG physically. The old Y74 / 20%-chord value is a geometric starting point, not an approved stability range.

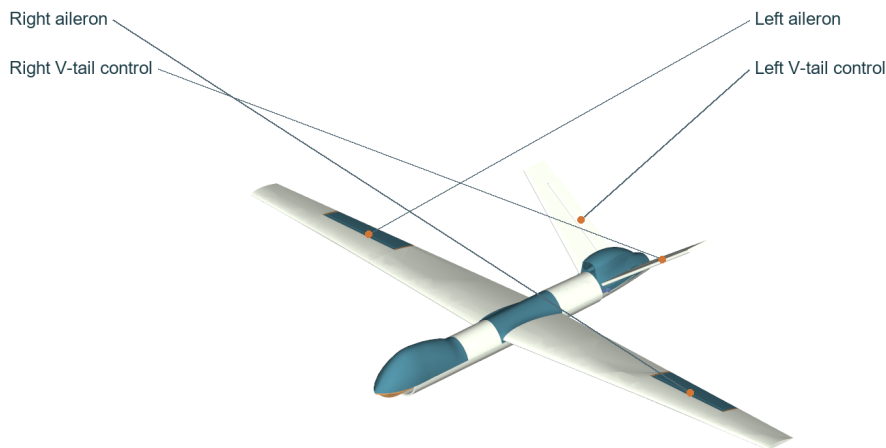
Choose any provisional balance target with an experienced aircraft builder from the actual final geometry and mass distribution. Moving the battery must preserve real frame support, restraint and removal access. A predicted balance point cannot approve a position that clashes with A34.

Agree progressive representative load tests for the keel and spar laps, roots/rods, reinforced plywood A44 servo supports, tail carrier, horns, motor mount and gear load path. Support the aircraft safely and record applied loads, deflection and recovery. Stop for cracks, slip, permanent bending or adhesive peel.

These drawings and mesh checks establish digital geometry only. No structural capacity, fatigue life or safe flight load is certified. Continue to powered propulsion testing only after the actual structure, restraint and controls are accepted.

Check before continuing: Gate stays closed until mass/CG, joint and control-load evidence are recorded and reviewed.

Physical acceptance gate 2: propulsion and ground



No flight-ready claim is made. The pictured aircraft remains an intended assembly until real acceptance is completed.

Qualify the exact EMAX ECO II 2807 1300KV, owned 9045 three-blade prop, 4S pack and ReadytoSky 40A ESC as one combination. The manufacturer recommends 6-7 inch propellers; a 9-inch prop requires actual current, thrust and temperature testing.

Use a secure purpose-made test restraint and competent help. Clear the complete propeller plane, fit the verified pusher prop/hub correctly and provide a safe disconnect. Record battery voltage sag, current, thrust, motor/ESC temperatures, vibration and mount movement progressively. Stop if the hardware limits are exceeded or any part loosens.

After qualified gear fitting, load the aircraft to its completed mass. Check propeller-ground clearance with gear deflection and nose-raised rotation, using the entire 228.6 mm prop disk. Inspect steering, tracking, braking/rolling behavior and attachment after controlled ground tests.

Thrust, launch method, runway requirement, endurance and safe control throws remain unestablished until tested and reviewed. Obtain experienced pilot acceptance of structure, CG, propulsion, controls, failsafe and site before any flight attempt.

Check before continuing: Flight remains gated until all physical evidence and pilot acceptance are documented. Passing a CAD or bench check alone does not open this gate.

If a check fails

Symptom	Stop and inspect	Safe next action
Wood tab will not seat	Orientation; measured thickness; documented relief	Compare actual mates; pare only specified local patches
Motor holes miss	Exact motor label; diameter19 PCD; actual mount	Template against real base; measure before any drilling
Servo buzz or binds	Horn/arm travel; pushrod rub; pin retention	Power off; fix mechanical cause; retest all directions
Board resets on servo motion	Vx/Vx2 voltage, BEC load and short	Prop off; isolate loads; correct supply/connection
Airspeed is negative/unstable	Ports, tube kink/leak, zero, exact sensor SKU	Identify actual ports/manual; test gentle positive flow
GPS/compass disturbed	Power wiring/metal distance; orientation	Reposition with measured clearance; recalibrate
Battery cannot be removed	Straps; nose/cover; hose and plug routing	Disconnect; revise service route before closing
Print/wood bond peels	Surface preparation, compatibility, cure	Remove load; remake tested joint; inspect all similar joints

Check before continuing: Record the failure and correction. Repeat every affected earlier check; do not continue because the cover hides it.

Foam ID to installed shape

Cut pattern	Installed CAD ID / plain name	Blank count
FW01 / FW02	A52 left / A49 right inner wing	Whole: 2; split ROOT/TIP: 4
FW03 / FW04	A38 right / A27 left outer wing	2
FW05 / FW06	A57 right / A22 left fixed V-tail	2
FW07	A56 + A17 forward/middle lower body	1
FW08	A42 aft lower body	1
FW09 / FW10	A08 forward / A15 aft upper covers	2
FW11	A20 tail underside	1
FW12 / FW13	A41 right / A60 left moving tail	2 per side, laminate to 10 mm

Check before continuing: Existing whole skins need no new seam. Already split skins need only their documented straight ROOT/TIP joins.

Print the missed aircraft pieces

Reuse sound prints and reprint missed or rejected parts. A11_2W_10GYR_SUP_L0p2_B8.stl is the left aileron; A21_2W_10GYR_SUP_L0p2_B8.stl is the right. Their upright orientation requires actual printer-height confirmation. A23 is the left fixed hinge rail, A36 the right. HTRB is the right tail-horn backing plate. These supplied STL geometries are unchanged.

A23/A36 filenames carry 3W_15GYR_NOSUP_L0p2_B8; each upright file is 16.676 x 19.699 x 270.204 mm. Confirm usable printer height, including any adhesion layers. HTRB_3W_100RECT_NOSUP_L0p2_B5.stl lies broad face down and is 22 x 24 x 1.5 mm.

Source settings recommend PLA+; the currently available spool is PLA. Use the actual filament-maker temperature range and a calibrated profile at 100% millimetre scale. PLA and PLA+ are not interchangeable strength ratings; qualify layer bonding, hinge/pin fit and cured joints on the actual prints.

Use 0.4 mm nozzle, 0.20 mm layers and initial 4 top/4 bottom solid layers. Inspect thin edges, supports, bores and complete skins in the slicer, print a fit coupon and reject cracked or warped parts. The plywood A44 ribs and A26 firewall are not printed in this cutting set.

IDs	Walls / infill	Supports / brim
A06 / A11 / A21	2 / 10% gyroid	Build plate only / 8 mm
A12	2 / 15% gyroid	Build plate only / 8 mm
A23 / A36	3 / 15% gyroid	Off / 8 mm
A45 / A55	4 / 35% gyroid	Off / 8 mm
HWL / HWR / HTLB / HTRB	3 / 100% rectilinear	Off / 5 mm
HTL / HTR	3 / 100% rectilinear	Build plate only / 5 mm

Record acceptance and keep the sources

Keep one build record: actual stock thickness, print masses, adhesive product/cure times, measured pin and servo-tab fits, motor pattern/thread engagement, voltage/current calibration, control throws, failsafe, structural loads, propulsion results, final mass/CG and pilot review.

Geometry and part identity come from the supplied source assembled CAD and R5 placed assembly, wood/print/foam records and removal solids. The original construction source is Michael Rehtin, Building a DIY REAPER Drone... Ended Badly. The current plywood ribs, rear mount and keel must remain paired with their current cutting set and local finishing details.

After every hard landing, ground strike or failed test inspect the high-load joints inside the access covers. Log the correction and repeat affected tests before an experienced pilot accepts the aircraft again.

Matek F405-WING V2 board and power

Matek M9N-5883 GPS / compass

Matek ASPD-4525 airspeed sensor

FlySky FS-IA10B receiver

ArduPilot Matek board and configuration

ArduPilot output functions

ArduPilot V-tail direction checks

ArduPilot airspeed setup

ArduPilot Plane failsafe

ArduPilot UART telemetry settings

EMAX ECO II 2807 official motor